EN.553.762 Nonlinear Optimization 2

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Chapter 1

Linear Programs

Definition 1.0.1. A linear program (LP) minimizes or maximizes a linear function

$$c^T x = \sum_{i=1}^{n} c_i x_i = \langle c, x \rangle$$

over a feasible region defined by linear inequalities $a_i^T x \leq b_i \forall i = 1, ..., m$:

We can also format this as:

$$\begin{cases} \min \max c^T x \\ s.t. \quad Ax \leq b \quad \text{(elementwise)} \end{cases}$$

where

- $A = \text{row}(a_m^T) = \text{col}(A_n)$
- $Ax = \text{col}(a_m^T x) = A_1 x_1 + ... + A_n x_n$
- n is the decision variable dimensions $x \in \mathbb{R}^n$
- m is the number of constraints $b \in \mathbb{R}^m$

Useful terms to know:

1. $\{x | a_i^T \leq b_i\}$ for fixed i is called a *half space*
2. $\mathcal{P} = \{x | a_i^T \leq b_i \forall i = 1, ..., m\}$ is called a *polyhedron*
3. If additional $\mathcal{P}$ is bounded, then it is called a *polytope*
4. We say a LP is *infeasible* if no x satisfies $Ax \leq b$, denoted by

$$\begin{cases} \min(\max) c^T x = \pm\infty \\ s.t. \quad Ax \leq b \end{cases}$$

5. We say a LP is *unbounded* if $\exists$ a sequence $x^{(i)} \in \mathbb{R}^n$ such that $\lim_{i\to\infty} c^T x^{(i)} = -\infty$.

6. We say $x \in \mathbb{R}^n$ is a *minimizer* (or maximizer) if $Ax \leq b$ and $c^T x^* \leq c^T x \forall x \text{s.t.} Ax \leq b$ (or $c^T x^* \geq c^T x \forall x \text{s.t.} Ax \leq b$)

7. Any $x \in \mathbb{R}^n$ is called a *solution*
   - x with $Ax \leq b$ is called a *feasible solution*
   - x minimizing/maximizing is called an *optimal solution*

Theorem 1.0.1. *Every LP is infeasible, unbounded, or has an optimal solution*

Proof. We prove this by counter-example. Consider a nonlinear program

$$\arg\min e^x \text{s.t.} x \in \mathbb{R}$$

. This is actually equivalent to taking an infinum, and results in non-attainment (We will worry about this for NLP but not LP) □

"Standard Form" LPs are

$$\begin{cases} \arg\min c^T x \quad \text{(always minimize)} \\ \text{s.t.} Ax = b \text{(equality constraints related to A)} \\ \quad x \geq 0 \text{(simple inequality constraints)} \end{cases}$$

Theorem 1.0.2. *Every LP can be equivalently rewritten in standard form*

Example 1: (Knapsack Problem) Suppose you have goods $i = 1, ..., n$ and each has three attributes $[w_i, v_i, p_i]$ for weight, volume, and payoff. Given weights and volume upper bounds W, V, you aim to maximize your payoff

$$\begin{cases} \arg\max p^T x = \sum p_i x_i \\ \text{s.t.} w^T x \leq W \\ \quad v^T x \leq V \\ \quad x \geq 0 \end{cases} \iff \begin{cases} \arg\min -p^T x \\ \text{s.t.} w^T x + s_1 \leq W \\ \quad v^T x + s_2 \leq V \\ \quad xs \geq 0 \end{cases}$$

Example 2: (Transportation Problem) Suppose we are shipping goods from supply centers $i = 1, ..., m$ to demand centers $j = 1, ..., n$. Each supply center has s_i supply and each demand center has d_j demand. Shipping cost from i to j is c_{ij} per unit

$$\begin{cases} \arg\min \sum_{ij} c_{ij} x_{ij} \\ \text{s.t.} \sum_j x_{ij} = s_i \forall i \\ \quad \sum_i x_{ij} = d_j \forall j \\ x \geq 0 \end{cases}$$

1.1 Extreme Points

Recall a typical picture of an LP's feasible region is $\mathcal{P} = \{x|Ax \leq b\}$ and the equivalent standard form is $\mathcal{P} = \{x|Ax = b, x \geq 0\}$, that is x is in a subspace and a non-negative orthant. Standard form gives a nice property that the feasible region is closed, i.e. $\{x|x_0 + \lambda d \forall \lambda \in \mathbb{R}\} \subsetneq \mathcal{P}$. We will also refer to extreme points as *corners*. There are three definitions:

- Geometric: we say $x \in \mathcal{P}$ is an *extreme point* if no $y, z \in \mathcal{P}$ with $x = \lambda y + (1-\lambda)z, \lambda \in [0,1]$ exists
- Optimization: we say $x \in \mathcal{P}$ is a *vertex* if some c has $c^T x < c^T y \forall y \in \mathcal{P}$
- Algebraic: we say $x \in \mathcal{P}$ is a *basic feasible solution (BFS)* if there exists a linearly independent a_i with $a_i^T x = b_i$ (implying x is the unique solution with those tight constraints)

Theorem 1.1.1. *For any polyhedron $\mathcal{P} = \{x|Ax \leq b\}$, a point $x \in \mathbb{R}^n$ is an extreme point iff it is a vertex and iff it is a BFS.*

Proof. We will prove this by showing Vertex $\implies$ Extreme Point $\implies$ BFS $\implies$ Vertex. Suppose x is a vertex. Consider any $y, z \in \mathcal{P}$. Now

$$\lambda(c^T x < c^T y) + (1-\lambda)(c^T x < c^T z) = c^T x \leq c^T(\lambda y + (1-\lambda)z)$$

$$\implies x \neq \lambda y + (1-\lambda)z \text{for any} \lambda \in [0,1]$$

Suppose x is not a BFS, Let $I \subseteq \{1, ..., m\}$ such that $a_i^T x = b_i$ ($i \notin I, a_i^T x < b_i$). Then $\{a_i\}_{i \in I}$ does not contain linearly independent vectors

$$\implies \exists d \neq 0 \quad a_i^T d = 0 \quad \forall i \in I$$

$$\implies a_i^T(x \pm \varepsilon d) = a_i^T x = b_i \quad \forall i \in I$$

$$\implies x \pm \varepsilon d \text{is feasible for some} \varepsilon \text{as small as we want}$$

Since $x = \frac{1}{2}((x + \varepsilon d) + (x - \varepsilon d))$, x is not an extreme point. Now suppose that x is BFS with I defined as before. Consider $c = -\sum_{i \in I} a_i$. All $y \in \mathcal{P}$ have

$$c^T y = -\sum_{i \in I} a_i^T y > -\sum_{i \in I} b_i = -\sum_{i \in I} a_i^T x = c^T x$$

BFS $\iff$ n linear independent $a_i^T x = b_i$ where x is an n-dimensional vector $\iff$ x is the only solution to $a_i^T y = b_i$ □

Recall a standard form polyhedron with m equalities and n more inequalities is $\mathcal{P} = \{x|Ax = b, x \geq 0\}$ (need $m < n$, assume A has independent rows).
Pick a *Basis* $B = \{B(1), ..., B(m)\} \subseteq \{1, ..., n\}$ and complement $\bar{B} = \{1, ..., n\}$. Let $x_B =$

$\vec{x}_{B(n)}$, $c_B = \vec{c}_{B(m)}$, $x_{\bar{B}} = \vec{x}_{B(n\times m)}$, $c_{\bar{B}} = \vec{c}_{B(n\times m)}$, $A_B = \text{col}(A_{B(m)})$, and $A_{\bar{B}} = \text{row}(A_{\bar{B}(n\times m)})$.
A BFS comes from forcing $x_{\bar{B}} = 0$

$$\begin{cases} Ax = b \\ x_{\bar{B}} = 0 \end{cases} \iff \begin{cases} A_B x_B + A_{\bar{B}} x_{\bar{B}} = b \\ x_{\bar{B}} = 0 \end{cases}$$

if and only if

$$\begin{cases} A_B x_B = b \\ x_{\bar{B}} = 0 \end{cases} \implies \begin{cases} x_B = A_B^{-1} b \\ x_{\bar{B}} = 0 \end{cases}$$

Lemma 1.1.1. *Every nonempty standard form polyhedron has a BFS*

Consider a standard form LP and recall that x^* is a minimizer if

$$\begin{cases} Ax^* = b, x^* \geq 0 \\ c^T x^* \leq c^T y \forall y \in \mathcal{P} \end{cases}$$

Theorem 1.1.2. *If an LP has a minimizer, then some BFS is a minimizer.*

Proof. Let x^* be a minimizer of $c^T x$ over $\mathcal{P}$. Define $Q = \{x | Ax = b, c^T x = c^T x^*, x \geq 0\}$. This means there must exist a x^* that is a BFS of Q, and we cannot write x^* as an average of minimizers (elements of Q). Consider a BFS x^* with basis B. This implies that a unique x^* solves

$$\begin{cases} Ax = b \\ x_{barB} = 0 \end{cases}$$

□

Consider a BFS $\bar{x}$ with basis B. Try increasing some $x_j = \varepsilon, j \in \bar{B}$.
Let $\tilde{x}$ uniquely solve

$$\begin{cases} Ax = b \\ x_j = \varepsilon \\ x_{\bar{B}} = 0 \end{cases} \iff \begin{cases} \tilde{x}_B = A_B^{-1} b - A_B^{-1} A_j \varepsilon \\ \tilde{x}_j = \varepsilon \\ \tilde{x}_{\bar{B}} = 0 \end{cases}$$

Primal feasibilitity is $x_B = A_B^{-1} b \geq 0$ and dual feasibility is $c - A^T y = c - A^T (A_B^{-1} c_B) \geq 0$

$$c^T \tilde{x} = \underbrace{c_B^T A_B^{-1} b}_{c^T \tilde{x}} - \underbrace{c_B^T A_B^{-1} A_j \epsilon + c_j \epsilon}_{\text{The reduced cost of j in basis B: } \bar{c}_j}$$

Theorem 1.1.3. *Consider a BFS $\bar{x}$ with B basis and $\bar{c}$ reduced costs. (i) if $\bar{c} \geq 0$, then $\bar{x}$ is a minimizer, and (ii) if $\bar{x}$ is a minimizer and nondegenerate, then $\bar{c} \geq 0$.*

Proof of (i). Suppose $\bar{c} \geq 0$. Consider any $x \in P = \{x \mid Ax = b,\ x \geq 0\}$.

We want to show:

$$c^T \bar{x} \leq c^T x.$$

Let

$$d = x - \bar{x}.$$

Since $Ax = b$ and $A\bar{x} = b$, we have

$$Ad = 0.$$

Partitioning into basic and nonbasic components:

$$A_B d_B + A_{\bar{B}} d_{\bar{B}} = 0.$$

Thus,

$$d_B = -A_B^{-1} A_{\bar{B}} d_{\bar{B}} = -\sum_{j \in \bar{B}} A_B^{-1}(A_j d_j).$$

Now,

$$c^T(x - \bar{x}) = c^T d = c_B^T d_B + c_{\bar{B}}^T d_{\bar{B}}.$$

Substitute for d_B:

$$= c_B^T \left(-\sum_{j \in \bar{B}} A_B^{-1} A_j d_j \right) + \sum_{j \in \bar{B}} c_j d_j.$$

$$= -\sum_{j \in \bar{B}} \left(c_B^T A_B^{-1} A_j - c_j \right) d_j.$$

$$= \sum_{j \in \bar{B}} \bar{c}_j \, d_j.$$

Since $x_j \geq 0$ and $\bar{x}_j = 0$ for $j \in \bar{B}$, we have $d_j \geq 0$, and by assumption $\bar{c}_j \geq 0$. Therefore,

$$\sum_{j \in \bar{B}} \bar{c}_j \, d_j \geq 0.$$

Hence,

$$c^T(x - \bar{x}) \geq 0 \quad \Rightarrow \quad c^T \bar{x} \leq c^T x.$$

□

Consider a primal standard form LP:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

Write

$$A = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix}.$$

Pick multipliers $y \in \mathbb{R}^m$. Then

$$\sum_{i=1}^{m} y_i \left(a_i^T x = b_i\right) \quad \Longleftrightarrow \quad y^T A x = y^T b.$$

Ensure that

$$y^T A \leq c^T \quad \text{(elementwise)}.$$

Since $x \geq 0$, we obtain

$$c^T x \geq y^T b.$$

The strongest lower bound is given by the dual LP:

$$\begin{aligned} \max \quad & b^T y \\ \text{s.t.} \quad & A^T y \leq c. \end{aligned}$$

Worked Example: A Two-Variable Production LP

Consider

$$\max_{x_1, x_2 \geq 0} 3x_1 + 2x_2 \quad \text{subject to} \quad x_1 + x_2 \leq 4, qquad 2x_1 + x_2 \leq 5. \tag{1.1.1}$$

The feasible vertices are $(0,0)$, $(0,4)$, $(5/2,0)$, and the intersection of the two active resource constraints, $(1,3)$. Their objective values are 0, 8, 15/2, and 9, respectively, so $x^\star = (1,3)$.

The same conclusion follows from a normal-cone calculation. At $(1,3)$ the active outward normals are $a_1 = (1,1)^T$ and $a_2 = (2,1)^T$. The objective gradient $c = (3,2)^T$ decomposes as

$$c = a_1 + a_2. \tag{1.1.2}$$

Hence c lies in the normal cone generated by the active constraints, which is the first-order optimality condition for maximizing a linear function over the polytope. This example illustrates three equivalent LP viewpoints: enumeration of extreme points, supporting hyperplanes, and complementary dual prices.

1.2 Duality

Theorem 1.2.1. *(Weak Duality) For any A, b, c,*

$$\begin{array}{lll} \min & c^T x & \\ s.t. & Ax = b & \geq \\ & x \geq 0 & \end{array} \quad \begin{array}{ll} \max & b^T y \\ s.t. & A^T y \leq c. \end{array}$$

Recall for a basis B, the reduced costs are

$$\bar{c}^T = c^T - c_B^T A_B^{-1} A \geq 0.$$

Define

$$y^T = c_B^T A_B^{-1}.$$

Then

$$A^T y \leq c.$$

Duality allow for several important things in optimization:

- Optimizes proofs
- Variational changes (aka shadow prices) via derivative w.r.t. b
- Lagrange multipliers as a penalty for infeasibility
- Changing points (Fenchel) and linear functions (gradients)

Theorem 1.2.2 (Strong Duality)**.** *For any (A, b, c), if at least one of the primal or dual LP is feasible, then*

$$p^* = \begin{array}{lll} \min & c^T x & \\ s.t. & Ax = b & = \\ & x \geq 0 & \end{array} \quad \begin{array}{ll} \max & b^T y \\ s.t. & A^T y \leq c \end{array} = d^*.$$

Proof. By weak duality,

$$p^* \geq d^*.$$

If the primal is unbounded, then $p^* = -\infty$, and the dual is infeasible, so $d^* = -\infty$.

Similarly, if the dual is unbounded, then the primal is infeasible, and

$$p^* = \infty = d^*.$$

□

Proof. WLOG assume the primal is feasible and not unbounded, so an optimal solution x^* exists.

We want to find a dual feasible y^* such that

$$c^T x^* = b^T y^*.$$

Construct a basis B such that:

- The associated primal solution $\bar{x}$ is feasible:

$$A\bar{x} = b, \quad \bar{x}_{\bar{B}} = 0, \quad \bar{x}_B = A_B^{-1} b \geq 0.$$

- The associated dual solution

$$y = A_B^{-T} c_B$$

  is feasible.

Dual feasibility follows since the reduced costs satisfy

$$\bar{c}^T = c^T - c_B^T A_B^{-1} A \geq 0 \quad \iff \quad A^T y \leq c.$$

Finally,

$$c^T \bar{x} = c_B^T \bar{x}_B = c_B^T A_B^{-1} b = b^T A_B^{-T} c_B = b^T y.$$

Thus $c^T \bar{x} = b^T y$, completing the proof. □

1.3 The Simplex Method

The core step is *pivoting*. Given a BFS $\bar{x}$ (nondegenerate) with basis B, so that

$$\bar{x}_B = A_B^{-1} b \geq 0, \quad \bar{x}_{\bar{B}} = 0,$$

fix $j \in \bar{B}$ and consider $\tilde{x}(\epsilon)$ defined by

$$\tilde{x}(\epsilon) = \bar{x} + \epsilon d,$$

where the direction d is given by

$$d = \begin{bmatrix} -A_B^{-1} A_j \\ 1 \\ 0 \end{bmatrix}.$$

Then $\tilde{x}(\epsilon)$ uniquely solves

$$Ax = b, \quad x_j = \epsilon, \quad x_{\bar{B} \setminus \{j\}} = 0.$$

1.3.1 Unbounded Case

If $d \geq 0$, then for all $\epsilon \geq 0$,

$$\tilde{x}(\epsilon) = \bar{x} + \epsilon d \geq 0,$$

and by construction,

$$A\tilde{x}(\epsilon) = b.$$

Thus $\tilde{x}(\epsilon)$ is feasible for all $\epsilon \geq 0$.

If $\bar{c}_j < 0$, then

$$c^T \tilde{x}(\epsilon) = c^T \bar{x} + \bar{c}_j \epsilon \to -\infty \quad \text{as } \epsilon \to \infty.$$

Hence the problem is unbounded.

1.3.2 Bounded Case

Suppose $\bar{c}_j < 0$ but $d \ngeq 0$. Then there exists some i such that

$$d_i < 0.$$

Feasibility requires

$$\tilde{x}_i(\epsilon) = \bar{x}_i + \epsilon d_i \geq 0,$$

which holds if and only if

$$\epsilon \leq \frac{\bar{x}_i}{-d_i}.$$

Define the step size

$$\epsilon^* = \min \left\{ \frac{\bar{x}_i}{-d_i} \;\middle|\; d_i < 0 \right\}.$$

Let

$$i \in \arg\min \left\{ \frac{\bar{x}_i}{-d_i} \;\middle|\; d_i < 0 \right\}.$$

Then $\tilde{x}(\epsilon^*)$ is feasible and defines the next BFS.

Lemma 1.3.1. *Let $B' = B \cup \{j\} \setminus \{i\}$. Then B' is a basis for $\tilde{x}(\epsilon^*)$. In particular, $\tilde{x}(\epsilon^*)$ is a BFS.*

Proof. Clearly, $\tilde{x}(\epsilon^*)$ satisfies

$$Ax = b, \quad x_{\bar{B}'} = 0.$$

Thus we need to show that $A_{B'}$ is invertible.

Note that $A_{B'}$ is obtained from A_B by replacing the i-th column with A_j. Let k be the position of i in B. Then

$$A_{B'} = \begin{bmatrix} A_{B(1)} & \cdots & A_j & \cdots & A_{B(m)} \end{bmatrix} = A_B + (A_j - A_i) e_k^T.$$

By the Sherman–Morrison formula, $A_{B'}$ is invertible if and only if

$$1 + e_k^T A_B^{-1}(A_j - A_i) \neq 0.$$

Compute:

$$e_k^T A_B^{-1} A_j = -d_k \quad \text{(from the definition of } d\text{)},$$

and

$$e_k^T A_B^{-1} A_i = e_k^T e_k = 1.$$

Hence,

$$1 + e_k^T A_B^{-1}(A_j - A_i) = 1 + (-d_k - 1) = -d_k.$$

Since $d_k \neq 0$ (indeed $d_k < 0$ by construction), we conclude that $A_{B'}$ is invertible. □

Before the 1980s, people used the Simplex Method, and it could be exponential time in the number of systems to be solved. Afterwards, Interior point methods were invented which solved these systems in polynomial time. Nowadays, we use First-Order approaches which instead of solving a system, it does a ton of matrix-vector multiplications

1.4 A Bit on LPs

The simplex method assumes that a basic feasible solution (BFS) is given.

Consider the standard form LP:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

To obtain an initial BFS, introduce slack variables and solve the Phase I problem:

$$\begin{aligned} \min \quad & \sum_i s_i \\ \text{s.t.} \quad & Ax + Is = b \\ & x, s \geq 0, \end{aligned}$$

where we assume $b \geq 0$. Then $(x, s) = (0, b)$ is a BFS.

Phase I and Phase II

- **Phase I:** Find a feasible solution by minimizing the sum of slack variables.
- **Phase II:** Use the BFS from Phase I to solve the original LP.

Adding constraints to the primal can break primal feasibility, while adding variables to the dual does not affect dual feasibility.

Worked Example: One Simplex Pivot

Write

$$\max 3x_1 + 2x_2 \quad \text{subject to} \quad x_1 + x_2 + s_1 = 4, quad 2x_1 + x_2 + s_2 = 5, quad x, s \geq 0. \tag{1.4.1}$$

Starting from the slack basis gives $(x_1, x_2, s_1, s_2) = (0, 0, 4, 5)$. Increasing x_1 changes the basic variables to $s_1 = 4 - x_1$ and $s_2 = 5 - 2x_1$, so the ratio test permits $x_1 \leq \min\{4, 5/2\} = 5/2$ and s_2 leaves the basis. At the new vertex $(5/2, 0, 3/2, 0)$, solve the second equation for $x_1 = (5 - x_2 - s_2)/2$. Substitution into the objective gives

$$3x_1 + 2x_2 = \frac{15}{2} + \frac{1}{2}x_2 - \frac{3}{2}s_2. \tag{1.4.2}$$

Thus x_2 enters. The first constraint becomes $s_1 = 3/2 - x_2/2 + s_2/2$, while $x_1 = 5/2 - x_2/2 - s_2/2$; the ratio test gives $x_2 = 3$. The resulting point $(x_1, x_2) = (1, 3)$ has objective 9. Both nonbasic reduced costs are then nonpositive, certifying optimality.

1.5 Modeling Choices

Consider the set

$$P = \{x \mid \|x\|_1 \leq 1\} = \left\{ x \;\middle|\; \sum_{i=1}^{n} |x_i| \leq 1 \right\}.$$

This polytope has:

- 2^n faces corresponding to inequalities

$$(\pm 1, \pm 1, \ldots, \pm 1)^T x \leq 1,$$

- $2n$ BFS given by vectors of the form

$$(0, \ldots, 0, \pm 1, 0, \ldots, 0).$$

Moreover,

$$P = \operatorname{conv}\left(\{\pm e_i\}_{i=1}^n\right) = \left\{ \sum_i \lambda_i x_i \;\middle|\; \sum_i \lambda_i = 1,\ \lambda_i \geq 0 \right\}.$$

Quadratic Formulation and Lifting Consider the problem:

$$\begin{aligned} \max \quad & x^T A x = \sum_{i,j} A_{ij} x_i x_j \\ \text{s.t.} \quad & x \in \{\pm 1\}^n. \end{aligned}$$

This can be written using matrix inner products:

$$\max \quad \langle A, X \rangle \quad \text{s.t.} \quad X = xx^T,\ x \in \{\pm 1\}^n,$$

where

$$\langle A, X \rangle = \mathrm{tr}(A^T X) = \sum_{i,j} A_{ij} X_{ij}.$$

Relaxing this, we obtain:

$$\max \quad \langle A, X \rangle \quad \text{s.t.} \quad X \in \mathrm{conv}\{xx^T \mid x \in \{\pm 1\}^n\}.$$

This leads to semidefinite programming relaxations, which can often recover good approximations (e.g., ~ 0.878 for Max-Cut).

Chapter 2

Beyond Linearity

We consider variables x living in a Euclidean, finite-dimensional space E equipped with an inner product $\langle \cdot, \cdot \rangle$ and norm

$$\|x\|^2 = \langle x, x \rangle.$$

Examples:

$$E = \mathbb{R}^n, \quad \langle x, y \rangle = x^T y,$$
$$E = \mathbb{R}^{n \times n}, \quad \langle X, Y \rangle = \operatorname{tr}(X^T Y).$$

We are interested in problems of the form

$$\begin{array}{ll} \min & f(x) \\ \text{s.t.} & x \in Q, \end{array}$$

or equivalently,

$$\begin{array}{ll} \min & f(x) \\ \text{s.t.} & g_i(x) \leq 0, \quad i = 1, \ldots, m, \end{array}$$

where $f, g_i : E \to \mathbb{R} \cup \{+\infty\}$.

The domain of a function $h : E \to \mathbb{R} \cup \{+\infty\}$ is

$$\operatorname{dom}(h) = \{x \in E \mid h(x) < +\infty\}.$$

2.1 Calculus

We say h is (Fréchet) differentiable at $x \in \operatorname{dom}(h)$ if there exists $\nabla h(x) \in E$ such that

$$\lim_{\Delta \to 0} \frac{h(x + \Delta) - \Big(h(x) + \langle \nabla h(x), \Delta \rangle\Big)}{\|\Delta\|} = 0.$$

The vector $\nabla h(x)$ is called the gradient.

If ∇h exists everywhere in $\mathrm{dom}(h)$ and is continuous, then $h \in C^1$.

We say h is twice (Fréchet) differentiable at x if there exists a linear operator

$$\nabla^2 h(x) : E \to E$$

such that

$$\lim_{\Delta \to 0} \frac{\|\nabla h(x+\Delta) - (\nabla h(x) + \nabla^2 h(x)(\Delta))\|}{\|\Delta\|} = 0.$$

The operator $\nabla^2 h(x)$ is called the Hessian. If it exists and is continuous, then $h \in C^2$.

2.2 Convexity

A function f is linear if

$$f(\lambda x) = \lambda f(x), \quad f(x+y) = f(x) + f(y), \quad \forall x, y \in E,\ \lambda \in \mathbb{R}.$$

It is affine if

$$f(\lambda x + (1-\lambda)y) = \lambda f(x) + (1-\lambda) f(y), \quad \forall x, y \in E,\ \lambda \in [0,1].$$

A function f is convex if

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y), \quad \forall x, y \in E,\ \lambda \in [0,1].$$

It is strictly convex if

$$f(\lambda x + (1-\lambda)y) < \lambda f(x) + (1-\lambda) f(y), \quad \forall x \neq y,\ \lambda \in (0,1).$$

A function is concave if the inequality is reversed.

A set $Q \subseteq E$ is convex if

$$\lambda x + (1-\lambda)y \in Q, \quad \forall x, y \in Q,\ \lambda \in [0,1].$$

Example

$$f(x) = |x| - 1$$

is convex but not strictly convex.

2.3 Families of Optimization Problems

2.3.1 Quadratic Programs (QP)

Let $E = \mathbb{R}^n$. Consider $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, and $H \in \mathbb{R}^{n\times n}$ symmetric positive semidefinite.

We study:

$$\begin{aligned} \min \quad & c^T x + \tfrac{1}{2} x^T H x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

If $H \succeq 0$, then:

$$v^T H v \geq 0 \quad \forall v \in \mathbb{R}^n \quad \iff \quad H = G^T G,$$

and the objective is convex.

Example 1 (Projection) Let $P = \{x \mid Ax = b,\ x \geq 0\}$. Given $\bar{x} \notin P$, the projection onto P solves:

$$\min_{x \in P} \tfrac{1}{2}\|x - \bar{x}\|_2^2.$$

Expanding,

$$\tfrac{1}{2}\|x\|^2 - \bar{x}^T x + \tfrac{1}{2}\|\bar{x}\|^2,$$

which is a QP with $H = I$, $c = -\bar{x}$ (up to constants).

Example 2 (LASSO)

$$\begin{aligned} \min \quad & \tfrac{1}{2}\|Ax - b\|_2^2 \\ \text{s.t.} \quad & \|x\|_1 \leq \lambda. \end{aligned}$$

Equivalently,

$$\tfrac{1}{2} x^T A^T A x - b^T A x + \tfrac{1}{2}\|b\|^2,$$

with $H = A^T A$.

2.3.2 Second-Order Cone Programs (SOCP)

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & \|A_i x - b_i\|_2 \leq f_i^T x + d_i, \quad i = 1, \ldots, k. \end{aligned}$$

LPs are included by setting $A_i = 0$, $b_i = 0$.

QPs can be embedded via epigraph reformulation:

$$\min t \quad \text{s.t.} \quad t \geq c^T x + \tfrac{1}{2} x^T H x.$$

Key representation:

$$\|y\|_2 \leq t \quad \iff \quad (y, t) \in \mathcal{K}_{\text{soc}}.$$

Example (LASSO revisited)

$$\min \|x\|_1 \quad \text{s.t.} \quad \|Ax - b\|_2 \leq \gamma,$$

which can be written as an SOCP using standard transformations.

2.3.3 Semidefinite Programs (SDP)

Let $E = \mathbb{S}^n$ be the space of symmetric matrices with inner product

$$\langle X, Y \rangle = \operatorname{tr}(X^T Y).$$

$$\begin{aligned} \min \quad & \langle C, X \rangle \\ \text{s.t.} \quad & \mathcal{A}(X) = b \\ & X \succeq 0, \end{aligned}$$

where $\mathcal{A} : \mathbb{S}^n \to \mathbb{R}^m$ is linear.

$X \succeq 0$ means X is positive semidefinite.

Connections

- If X is diagonal, SDP reduces to LP.
- SOCP constraints satisfy:

$$\|x\|_2 \leq t \quad \Longleftrightarrow \quad \begin{bmatrix} tI & x \\ x^T & t \end{bmatrix} \succeq 0.$$

Example: Max-Cut Relaxation

The Max-Cut problem:

$$\min x^T A x \quad \text{s.t.} \quad x \in \{\pm 1\}^n.$$

Lift to matrix form:

$$X = x x^T.$$

Relaxation:

$$\begin{aligned} \min \quad & \langle A, X \rangle \\ \text{s.t.} \quad & \operatorname{diag}(X) = \mathbf{1} \\ & X \succeq 0, \end{aligned}$$

dropping the rank constraint.

This is the Goemans–Williamson SDP relaxation.

2.3.4 Conic Programs

A conic program is

$$\begin{aligned} \min \quad & \langle c, x \rangle \\ \text{s.t.} \quad & Ax = b \\ & x \in K, \end{aligned}$$

for some $c \in E$, where $A : E \to \mathbb{R}^m$ is linear, and K is a closed, convex cone in E.

The cone of copositive matrices:

$$K^{\text{cop}} = \{X \in \mathbb{R}^{n \times n} \mid v^T X v \geq 0 \ \ \forall v \geq 0\}.$$

Checking $X \in K^{\text{cop}}$ is NP-hard.

(Solve with interior point methods $\to$ tells us which K are tractable.)

2.3.5 Convex Programs

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \ldots, m, \end{aligned}$$

where f, g_i are closed, convex, proper (c.c.p.).

f is closed if

$$\liminf_{x' \to x} f(x') = f(x), \quad \forall x \in E.$$

f is proper if

$$\text{dom}(f) = \{x \mid f(x) \in \mathbb{R}\} \text{ is nonempty.}$$

Given any closed, convex, nonempty feasible region

$$S = \{x \mid Ax = b, \ x \in K\},$$

claim:

$$\text{dist}_S(x) = \inf\{\|x - x'\| \mid x' \in S\}$$

is c.c.p.

$$\begin{aligned} \min \quad & \langle c, x \rangle \\ \text{s.t.} \quad & x \in S \end{aligned} \iff \begin{aligned} \min \quad & \langle c, x \rangle \\ \text{s.t.} \quad & \text{dist}_S(x) \leq 0. \end{aligned}$$

Epigraph Reformulation

$$\begin{array}{ll}\min & f(x) \\ \text{s.t.} & g_i(x) \le 0\end{array} \quad = \quad \min_{x \in E} h(x),$$

where

$$h(x) = \begin{cases} f(x) & \text{if } x \text{ is feasible,} \\ +\infty & \text{otherwise.} \end{cases}$$

$$= \begin{array}{ll}\min & t \\ \text{s.t.} & (x,t) \in \operatorname{epi}(h)\end{array}$$

where

$$\operatorname{epi}(h) = \{(x,t) \mid h(x) \le t\}.$$

$$= \begin{array}{ll}\min & t \\ \text{s.t.} & (x,t,z) \in K^{\text{epi}} \\ & z = 1\end{array}$$

where

$$K^{\text{epi}} = \left\{ (x,t,z) \;\middle|\; \left(\frac{x}{z}, \frac{t}{z}\right) \in \operatorname{epi}(h) \right\}.$$

Thus:

- linear objective
- linear constraints
- one cone constraint

$$\Longrightarrow \begin{array}{ll}\min & \langle c, x \rangle \\ \text{s.t.} & x \in K \\ & Ax = b.\end{array}$$

Worked Example: Smallest Enclosing Ball as an SOCP

Given points $a_1, \ldots, a_N \in \mathbb{R}^d$, the smallest enclosing ball solves

$$\min_{c,r} \; r \quad \text{subject to} \quad \|c - a_i\|_2 \le r, qquadi = 1, \ldots, N. \tag{2.3.1}$$

Every constraint is second-order conic. For the one-dimensional points $a_1 = -2$, $a_2 = 1$, and $a_3 = 4$, the constraints reduce to $|c+2| \le r$, $|c-1| \le r$, and $|c-4| \le r$. Any enclosing interval must cover the two extremes, so $2r \ge 6$. Equality is achieved by $c = 1$, $r = 3$. The middle observation is inactive. In higher dimensions, active boundary points play the same role and their conic multipliers balance to produce the center stationarity condition.

2.4 Optimality Conditions

Consider

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in Q \subseteq E. \end{aligned}$$

Definitions

$\bar{x} \in E$ is a *local minimizer* if

$$f(x) \geq f(\bar{x}) \quad \forall x \text{ in a neighborhood of } \bar{x} \text{ and feasible.}$$

$\bar{x} \in E$ is a *global minimizer* if

$$f(x) \geq f(\bar{x}) \quad \forall x \in Q.$$

Directional Derivative

Define the directional derivative of f at $\bar{x}$ in direction d:

$$f'(\bar{x}; d) = \lim_{t \to 0^+} \frac{f(\bar{x} + td) - f(\bar{x})}{t}.$$

If $f \in C^1$, then

$$f'(\bar{x}; d) = \langle \nabla f(\bar{x}), d \rangle.$$

Example For $f(x) = |x|$ at $\bar{x} = 0$:

$$f'(0; 1) = 1, \quad f'(0; -1) = -1.$$

Normal Cone

Define the normal cone of Q at $\bar{x}$:

$$N_Q(\bar{x}) = \{d \mid \langle d, x - \bar{x} \rangle \leq 0 \ \ \forall x \in Q\}.$$

2.4.1 First-Order Necessary Condition

Theorem 2.4.1. *Suppose Q is closed and convex and $\bar{x} \in Q$ is a local minimizer of f over Q.*

If for any $x \in Q$, $f'(\bar{x}; x - \bar{x})$ exists, then

$$f'(\bar{x}; x - \bar{x}) \geq 0.$$

If f is differentiable at $\bar{x}$, then

$$\langle \nabla f(\bar{x}), x - \bar{x} \rangle \geq 0 \quad \forall x \in Q,$$

equivalently,

$$-\nabla f(\bar{x}) \in N_Q(\bar{x}).$$

Proof. Suppose to the contrary that for some $x \in Q$,

$$f'(\bar{x}; x - \bar{x}) < 0.$$

Then for some $0 < t < \delta$,

$$\frac{f(\bar{x} + t(x - \bar{x})) - f(\bar{x})}{t} < 0 \;\Rightarrow\; f(\bar{x} + t(x - \bar{x})) < f(\bar{x}).$$

But

$$\bar{x} + t(x - \bar{x}) = tx + (1 - t)\bar{x} \in Q$$

by convexity, contradicting local optimality. □

2.4.2 First-Order Sufficient Condition

Theorem 2.4.2. *Suppose f and Q are both closed and convex.*

If for all $x \in Q$,

$$f'(\bar{x}; x - \bar{x}) \geq 0$$

for some fixed $\bar{x} \in Q$, then $\bar{x}$ is a global minimizer.

In particular, if f is differentiable at $\bar{x}$,

$$-\nabla f(\bar{x}) \in N_Q(\bar{x}) \iff \bar{x} \text{ is a global minimizer.}$$

Proof. The key claim is that convexity implies

$$t \mapsto \frac{f(\bar{x} + t(x - \bar{x})) - f(\bar{x})}{t}$$

is nondecreasing for $t > 0$.

Thus for any $x \in Q$,

$$\lim_{t \to 0^+} \frac{f(\bar{x} + t(x - \bar{x})) - f(\bar{x})}{t} \geq 0.$$

Taking $t = 1$,

$$f(x) - f(\bar{x}) \geq 0 \;\Rightarrow\; f(x) \geq f(\bar{x}),$$

so $\bar{x}$ is globally optimal. □

Worked Example: Projection onto the Probability Simplex

The Euclidean projection of $v \in \mathbb{R}^n$ onto $\Delta_n = \{x \geq 0 : \mathbf{1}^T x = 1\}$ solves

$$\min_x \frac{1}{2} \|x - v\|_2^2 \quad \text{subject to} \quad x \geq 0, \quad \mathbf{1}^T x = 1. \tag{2.4.1}$$

With multiplier τ for the equality and multipliers $\mu_i \geq 0$ for nonnegativity, stationarity gives $x_i - v_i + \tau - \mu_i = 0$. Complementarity therefore yields

$$x_i = (v_i - \tau)_+, \qquad \sum_{i=1}^{n} (v_i - \tau)_+ = 1. \tag{2.4.2}$$

The left side of the scalar equation is continuous and strictly decreasing on the relevant interval, so τ is unique. For $v = (0.2, -0.1, 1.4)$, only the first and third coordinates survive. Solving $(0.2 - \tau) + (1.4 - \tau) = 1$ gives $\tau = 0.3$, but this would make the first coordinate negative, so the assumed active set is inconsistent. Retaining only the third coordinate gives $\tau = 0.4$ and $\Pi_{\Delta_3}(v) = (0, 0, 1)$. The active-set check is essential.

2.5 Algorithms

2.5.1 Frank–Wolfe Method

Given

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in Q, \end{aligned}$$

approximate by the linearized problem:

$$\begin{aligned} \min \quad & f(x_k) + \langle \nabla f(x_k), x - x_k \rangle \\ \text{s.t.} \quad & x \in Q. \end{aligned}$$

If Q is polyhedral, run simplex or enumerate BFS.

If $Q = B(0, 1)$, the solution is

$$\bar{x} = \frac{-\nabla f(x_k)}{\|\nabla f(x_k)\|_2}.$$

Iteration:

$$\bar{x}_{k+1} = \arg\min_{x \in Q} \langle \nabla f(x_k), x \rangle,$$

$$x_{k+1} = x_k + \beta_k (\bar{x}_{k+1} - x_k).$$

Choose $\beta_k = \frac{1}{\sqrt{k+1}}$ or via line search:

$$x_{k+1} \in \arg\min_{x \in [x_k, \bar{x}_{k+1}]} f(x).$$

Define

$$G_k = \langle \nabla f(x_k), x_k - \bar{x}_{k+1} \rangle \geq 0.$$

Then

$$f(x_{k+1}) = f(x_k + \beta_k (\bar{x}_{k+1} - x_k))$$

$$\leq f(x_k) + \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{L}{2}\|x_{k+1} - x_k\|_2^2$$
$$= f(x_k) - \beta_k G_k + \frac{L}{2}\beta_k^2 \|\bar{x}_{k+1} - x_k\|_2^2.$$

Assuming $\|\bar{x}_{k+1} - x_k\| \leq 2D$:

$$f(x_{k+1}) \leq f(x_k) - \beta_k G_k + 2LD^2\beta_k^2.$$

Summing:

$$f(x_0) - \min_{x \in Q} f(x) \geq \sum_{k=1}^{T} \beta_k G_k - 2LD^2 \sum_{k=1}^{T} \beta_k^2.$$

Thus,

$$\min_k G_k \leq \frac{f(x_0) - p^* + 2LD^2 \log T}{\sqrt{T+1}}.$$

2.5.2 Projected Gradient Descent

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in Q. \end{aligned}$$

Iteration:

$$x_{k+1} = \text{proj}_Q(x_k - \beta_k \nabla f(x_k)),$$

where

$$\text{proj}_Q(y) = \arg\min_{z \in Q} \frac{1}{2}\|z - y\|^2.$$

Lemma 2.5.1. *If $z = \text{proj}_Q(y)$, then*

$$y - z \in N_Q(z).$$

Proof.

$$-\nabla \left(\tfrac{1}{2}\|z - y\|^2 \right) \in N_Q(z) \quad \Rightarrow \quad -(z - y) \in N_Q(z).$$

□

Thus,

$$n_k = (x_k - \beta_k \nabla f(x_k)) - x_{k+1} \in N_Q(x_{k+1}),$$

i.e.

$$-\nabla f(x_k) + \frac{1}{\beta_k}(x_k - x_{k+1}) \in N_Q(x_{k+1}).$$

If $x_k \to \bar{x}$, then

$$-\nabla f(\bar{x}) \in N_Q(\bar{x}).$$

2.5.3 Proximal Point Method

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in Q. \end{aligned}$$

Iteration:

$$x_{k+1} \in \arg\min_{x \in Q} \left\{ f(x) + \frac{\rho}{2}\|x - x_k\|^2 \right\}.$$

Optimality condition:

$$-\nabla \left(f(x) + \tfrac{\rho}{2}\|x - x_k\|^2 \right) \Big|_{x=x_{k+1}} \in N_Q(x_{k+1}),$$

i.e.

$$-\nabla f(x_{k+1}) - \rho(x_{k+1} - x_k) \in N_Q(x_{k+1}).$$

2.6 Constrained Optimality Conditions

Theorem 2.6.1 (Weierstrass)**.** *For any closed nonempty set $Q \subseteq E$, and closed $f : Q \to \mathbb{R}$ with bounded level sets*

$$\{x \mid f(x) \leq \lambda\},$$

there exists a global minimizer of f over Q.

Theorem 2.6.2 (Basic Separating Hyperplane Theorem)**.** *For any closed convex $Q \subseteq E$, and $y \notin Q$, there exist $a \in E$, $b \in \mathbb{R}$ such that*

$$\langle a, y \rangle > b \geq \langle a, x \rangle \quad \forall x \in Q.$$

Proof. Consider $f(x) = \frac{1}{2}\|x - y\|_2^2$. Then f has bounded level sets.

By Weierstrass, there exists $\bar{x}$ minimizing f over Q, hence

$$-\nabla f(\bar{x}) \in N_Q(\bar{x}).$$

Since $\nabla f(\bar{x}) = \bar{x} - y$, we get

$$y - \bar{x} \in N_Q(\bar{x}),$$

i.e.

$$\langle y - \bar{x}, x - \bar{x} \rangle \leq 0 \quad \forall x \in Q.$$

Pick $a = y - \bar{x}$ and $b = \langle y - \bar{x}, \bar{x} \rangle$. Then

$$\langle a, y \rangle > b \geq \langle a, x \rangle \quad \forall x \in Q.$$

□

Theorem 2.6.3 (Small Gradients Exist). *If $f : E \to \mathbb{R}$ is C^1 and bounded below, then there exists a sequence $x_k \in E$ such that*

$$\nabla f(x_k) \to 0.$$

Proof. Fix $\epsilon > 0$ and define

$$h(x) = f(x) + \epsilon\|x\|.$$

Then h has bounded level sets, so it attains a global minimizer x_ϵ.

Let $d = -\nabla f(x_\epsilon)$. For small $t > 0$,

$$\frac{f(x_\epsilon + td) - f(x_\epsilon)}{t} = \frac{h(x_\epsilon + td) - h(x_\epsilon)}{t} - \epsilon\frac{\|x_\epsilon + td\| - \|x_\epsilon\|}{t}.$$

Thus

$$\langle \nabla f(x_\epsilon), d\rangle \geq -\epsilon\|d\|.$$

Since $d = -\nabla f(x_\epsilon)$,

$$-\|\nabla f(x_\epsilon)\|^2 \geq -\epsilon\|\nabla f(x_\epsilon)\| \quad \Rightarrow \quad \|\nabla f(x_\epsilon)\| \leq \epsilon.$$

□

Theorem 2.6.4 (Gordan's Theorem of Alternatives). *For any $a_1, \ldots, a_m \in E$, exactly one of the following holds:*

1. *$\exists \lambda \geq 0$, $\sum_i \lambda_i = 1$ such that*
$$\sum_{i=1}^m \lambda_i a_i = 0,$$
2. *$\exists x \in E$ such that*
$$\langle a_i, x\rangle < 0 \quad \forall i.$$

Lemma 2.6.1 (Farkas Lemma). *For any $a_1, \ldots, a_m \in E$, $c \in E$, exactly one holds:*

1. *$\exists \mu \geq 0$ such that*
$$\sum_{i=1}^m \mu_i a_i = c,$$
2. *$\exists x \in E$ such that*
$$\langle a_i, x\rangle \leq 0 \quad \forall i, \quad \langle c, x\rangle > 0.$$

Constrained Optimality Conditions

Consider

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0. \end{aligned}$$

Assume $f, g_i \in C^1$, and let $\bar{x}$ satisfy $g_i(\bar{x}) \leq 0$.

Define the active set:

$$I(\bar{x}) = \{i \mid g_i(\bar{x}) = 0\}.$$

Lagrangian

For $\lambda \geq 0$, define

$$L(x; \lambda) = f(x) + \sum_i \lambda_i g_i(x).$$

We say $\lambda \geq 0$ is a Lagrange multiplier vector at $\bar{x}$ if:

1. $\bar{x}$ is a critical point of $L(\cdot; \lambda)$:
$$\nabla_x L(\bar{x}; \lambda) = 0,$$
2. complementary slackness:
$$\lambda_i g_i(\bar{x}) = 0 \quad \forall i.$$

Theorem 2.6.5 (Fritz John)**.** *If $\bar{x}$ is a local minimizer, then there exists $(\lambda_0, \lambda) \in \mathbb{R}_+ \times \mathbb{R}^m_+$, not all zero, such that*

$$\lambda_0 \nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) = 0,$$

$$\lambda_i g_i(\bar{x}) = 0.$$

If $\lambda_0 > 0$, then λ/λ_0 is a Lagrange multiplier vector.

To guarantee $\lambda_0 \neq 0$, we require that there is no $\lambda \geq 0$, not all zero, such that

$$\sum_{i \in I(\bar{x})} \lambda_i \nabla g_i(\bar{x}) = 0.$$

By Gordan's theorem, this is equivalent to the existence of $d \in E$ such that

$$\langle \nabla g_i(\bar{x}), d \rangle < 0 \quad \forall i \in I(\bar{x}).$$

Such a d is a descent direction for all active constraints.

Worked Example: KKT Conditions on a Disk

Minimize $f(x) = x_1 + x_2$ subject to $g(x) = x_1^2 + x_2^2 - 1 \le 0$. The unconstrained linear objective has no minimizer, so the disk boundary must be active. The KKT equations are

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2\lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0, \qquad x_1^2 + x_2^2 = 1, qquad ad \lambda \ge 0. \tag{2.6.1}$$

Stationarity gives $x_1 = x_2 = -1/(2\lambda)$. Feasibility then yields $\lambda = 1/\sqrt{2}$ and $x^\star = (-1/\sqrt{2}, -1/\sqrt{2})$. Since the problem is convex and Slater's condition holds at the origin, these conditions are sufficient. Geometrically, the negative objective gradient is an outward normal to the disk at the solution.

Chapter 3

Duality

Theorem 3.0.1 (Fritz–John)**.** *If $\bar{x}$ is a local minimizer of*

$$\min f(x) \quad s.t.\ g_i(x) \leq 0 \ \ \forall i,$$

and f, g_i are differentiable at $\bar{x}$, then there exists $(\lambda_0, \lambda) \in \mathbb{R}_+ \times \mathbb{R}^m_+$, not all zero, such that

$$\lambda_0 \nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) = 0,$$

$$\lambda_i g_i(\bar{x}) = 0 \quad \forall i.$$

We want to guarantee $\lambda_0 \neq 0$.

$$\iff \nexists \lambda \geq 0,\ \lambda \neq 0 \text{ such that } \sum_i \lambda_i \nabla g_i(\bar{x}) = 0,\ \lambda_i g_i(\bar{x}) = 0.$$

$$\iff \nexists \lambda \geq 0,\ \sum_{i \in I(\bar{x})} \lambda_i = 1,\ \sum_{i \in I(\bar{x})} \lambda_i \nabla g_i(\bar{x}) = 0.$$

$$\iff \exists d \in E \text{ such that } \langle \nabla g_i(\bar{x}), d \rangle < 0 \quad \forall i \in I(\bar{x}).$$

Theorem 3.0.2 (Karush–Kuhn–Tucker)**.** *Under the same conditions as Fritz–John, if MFCQ holds at $\bar{x}$, then there exists $\lambda \geq 0$ such that*

$$\nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) = 0,$$

$$\lambda_i g_i(\bar{x}) = 0 \quad \forall i.$$

Proof. Fritz–John gives (λ_0, λ). MFCQ implies $\lambda_0 \neq 0$. Normalize to $(1, \lambda/\lambda_0)$. □

3.1 Mangasarian–Fromovitz Constraint Qualification (MFCQ)

Lemma 3.1.1. *For convex $g_i \in C^1$, MFCQ holds at all $\bar{x}$ feasible iff there exists x_s such that*

$$g_i(x_s) < 0 \quad \forall i.$$

Proof. If MFCQ holds at $\bar{x}$, then $\bar{x} + td$ is strictly feasible.

If x_s exists, then for any $\bar{x}$, let $d = x_s - \bar{x}$. For $t < 1$,

$$\frac{g_i(\bar{x} + td) - g_i(\bar{x})}{t} \leq \frac{g_i(x_s) - g_i(\bar{x})}{1} < 0.$$

□

Proof of Fritz–John. Let

$$h(x) = \max\{f(x) - f(\bar{x}),\ g_i(x)\}_{i \in I(\bar{x})}.$$

Then $\bar{x}$ is a local minimizer of h.

Claim:

$$h'(\bar{x}; d) = \max\{\langle \nabla f(\bar{x}), d\rangle,\ \langle \nabla g_i(\bar{x}), d\rangle\}_{i \in I(\bar{x})}.$$

First-order optimality gives

$$h'(\bar{x}; d) \geq 0 \quad \forall d,$$

i.e.

$$\nexists d \text{ such that } \begin{cases} \langle \nabla f(\bar{x}), d\rangle < 0, \\ \langle \nabla g_i(\bar{x}), d\rangle < 0. \end{cases}$$

Thus,

$$\exists \lambda_0, \lambda \geq 0 \text{ such that } \lambda_0 \nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) = 0.$$

□

3.2 Lagrangian and Duality

Recall the Lagrangian:

$$L(x; \lambda) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) = f(x) + \lambda^T g(x).$$

Claim:

$$\sup_{\lambda \geq 0} L(x; \lambda) = \begin{cases} f(x) & \text{if } g_i(x) \leq 0, \\ +\infty & \text{otherwise.} \end{cases}$$

Primal problem

$$p_* = \min f(x) \quad \text{s.t. } g_i(x) \leq 0 = \min_x \max_{\lambda \geq 0} L(x; \lambda).$$

Dual problem

$$d_* = \max_{\lambda \geq 0} \min_x L(x; \lambda) = \max_{\lambda \geq 0} \Phi(\lambda),$$

where $\Phi(\lambda)$ is the dual function.

Theorem 3.2.1 (Weak Duality)**.** *For any $f, g_i : E \to \mathbb{R} \cup \{+\infty\}$,*

$$p_* \geq d_*.$$

Proof. Let $\lambda^{(i)} \geq 0$ attaining d_*. For any feasible $\bar{x}$,

$$f(\bar{x}) = \sup_{\lambda \geq 0} L(\bar{x}; \lambda) \geq \limsup L(\bar{x}; \lambda^{(i)}) \geq \limsup \inf_x L(x; \lambda^{(i)}) = d_*.$$

□

3.3 Value Function and Strong Duality

Define the value function:

$$v(\Delta) = \min f(x) \quad \text{s.t. } g_i(x) \leq \Delta_i.$$

Theorem 3.3.1 (Lagrangian Strong Duality)**.** *Suppose $v : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$ is convex and $v(0) = p_*$. Then*

$$p_* = d_* \quad \iff \quad v \text{ is lower semicontinuous at } 0.$$

A function is lower semicontinuous at $\bar{x}$ if

$$\liminf_{x \to \bar{x}} f(x) \geq f(\bar{x}).$$

Notes: - If all f, g_i are convex, then $v(\cdot)$ is convex. - If there exists x_s with $g_i(x_s) < 0$ (Slater), then v is lower semicontinuous.

Theorem 3.3.2 (Strong Duality with Attainment)**.** *If $f, g_1, \ldots, g_m$ are convex and there exists x_s with $g_i(x_s) < 0$, then*

$$p_* = d_*$$

and the optimum is attained.

(Proof deferred: Fenchel duality)

Example: Linear Programming

$$\min -b^T y \quad \text{s.t. } A^T y - c \le 0$$

$$= \min_y \max_{x \ge 0} \left(-b^T y + x^T (A^T y - c) \right) \ge \max_{x \ge 0} \min_y (\cdot)$$

$$\Phi(x) = \min_y \left(-b^T y + x^T A^T y - x^T c \right) = -c^T x + \min_y (Ax - b)^T y$$

$$= \begin{cases} -c^T x & Ax = b, \\ -\infty & \text{otherwise.} \end{cases}$$

$$\Rightarrow \max_{x \ge 0} \Phi(x) = \max -c^T x \quad \text{s.t. } Ax = b,\ x \ge 0.$$

Worked Example: Constraint Qualification Failure

Consider

$$\min_x x \quad \text{subject to} \quad x^2 \le 0. \tag{3.3.1}$$

The only feasible point is $x^\star = 0$, hence it is optimal. However, $\nabla g(0) = 0$, so MFCQ fails: there is no direction d satisfying $\nabla g(0)^T d < 0$. KKT stationarity would require $1 + \lambda\, 2x^\star = 1 = 0$, which is impossible for every multiplier. The Fritz–John conditions remain valid with objective multiplier zero. This example shows why KKT necessity is a regularity theorem rather than a purely algebraic consequence of local optimality.

3.4 Norm Optimization

Given a norm $\|\cdot\|$, the dual norm is

$$\|v\|_* = \sup_{\|x\|=1} \langle v, x \rangle.$$

$$p_* = \min \|x\| \quad \text{s.t. } Ax \le b.$$

$$L(x; \lambda) = \|x\| + \lambda^T (Ax - b).$$

$$\Phi(\lambda) = \begin{cases} -\langle b, \lambda \rangle & \|A^T \lambda\|_* \le 1, \\ +\infty & \text{otherwise.} \end{cases}$$

$$d_* = \max -\langle b, \lambda \rangle \quad \text{s.t. } \|A^T \lambda\|_* \le 1,\ \lambda \ge 0.$$

3.5 Subgradient Calculus

Consider $f : E \to \mathbb{R} \cup \{+\infty\}$.

Convex:

$$\forall x, y \in E,\ \lambda \in [0,1], \quad f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y).$$

Proper:

$$\operatorname{dom} f = \{x \mid f(x) \in \mathbb{R}\} \neq \emptyset.$$

Closed: f is lower semicontinuous on $\operatorname{dom} f$, i.e.

$$\forall x \in \operatorname{dom} f, \quad \liminf_{x' \to x} f(x') \ge f(x).$$

3.5.1 Subgradients

Definition: We say $\phi \in E$ is a subgradient of f at $\bar{x} \in \operatorname{dom} f$ if

$$\forall x \in E, \quad f(x) \ge f(\bar{x}) + \langle \phi, x - \bar{x} \rangle.$$

Subdifferential:

$$\partial f(\bar{x}) = \{\phi \mid \forall x,\ f(x) \ge f(\bar{x}) + \langle \phi, x - \bar{x} \rangle\}.$$

3.5.2 Basic Properties

If f is C^1 and convex:

$$\partial f(x) = \{\nabla f(x)\}.$$

If

$$f(x) = \max_{i=1,\dots,m} f_i(x), \quad f_i \in C^1 \text{ convex},$$

then

$$\partial f(x) = \left\{ \sum_{i \in I(x)} \lambda_i \nabla f_i(x) \;\middle|\; \sum_{i \in I(x)} \lambda_i = 1,\ \lambda_i \ge 0 \right\},$$

where

$$I(x) = \{i \mid f_i(x) = f(x)\}.$$

Examples

$$f(x) = |x| = \max\{-x, x\}$$

$$\partial f(x) = \begin{cases} \{1\} & x > 0, \\ [-1,1] & x = 0, \\ \{-1\} & x < 0. \end{cases}$$

For a norm $f(x) = \|x\|$:

$$\phi \in \partial f(0) \iff \forall x,\ \|x\| \geq \langle \phi, x \rangle \iff \|\phi\|_* \leq 1.$$

$$\Rightarrow \partial \|\cdot\|(0) = \{\phi \mid \|\phi\|_* \leq 1\}.$$

3.5.3 Directional Derivatives

Lemma 3.5.1.

$$\phi \in \partial f(x) \iff \langle \phi, v \rangle \leq f'(x; v) \quad \forall v \in E.$$

Proof. Let $\phi \in \partial f(x)$.

$$f'(x; v) = \lim_{t \downarrow 0} \frac{f(x + tv) - f(x)}{t} \geq \lim_{t \downarrow 0} \frac{f(x + tv) - f(x)}{t} \geq \langle \phi, v \rangle.$$

□

By the lemma:

$$f'(x; v) \geq \max_{\phi \in \partial f(x)} \langle \phi, v \rangle.$$

We seek a subgradient attaining equality.

Existence of Subgradients

Let $S = \operatorname{epi} f$.

Pick $t_n \downarrow 0$ and define

$$x_n = \bar{x} + t_n v, \quad f_n = f(\bar{x}) + t_n f'(\bar{x}; v).$$

Then

$$f'(\bar{x}; v) \leq \frac{f(x_n) - f(\bar{x})}{t_n} \Rightarrow f(x_n) \geq f(\bar{x}) + t_n f'(\bar{x}; v).$$

Thus (x_n, f_n) lies on or below the graph.

By separating hyperplane theorem, there exists (g_n, h_n) such that

$$\langle (g_n, h_n), (x, t) - (x_n, f_n) \rangle \leq 0 \quad \forall (x, t) \in \operatorname{epi} f.$$

Rescale so $h_n = -1$.

Then

$$\langle (g_n, -1), (x, f(x)) - (x_n, f_n) \rangle \leq 0$$
$$\iff f(x) \geq f_n + \langle g_n, x - x_n \rangle.$$

Taking limits,

$$f(x) \geq f(\bar{x}) + \langle \bar{g}, x - \bar{x} \rangle \quad \forall x,$$

so

$$\bar{g} \in \partial f(\bar{x}).$$

Conclusion:

$$\partial f(\bar{x}) \neq \emptyset \quad \text{for all } \bar{x} \in \operatorname{int}(\operatorname{dom} f).$$

3.5.4 Max Formula

Theorem 3.5.1 (Max Formula)**.** *For any closed, convex, proper f with $\bar{x} \in \operatorname{int}(\operatorname{dom} f)$, and any $v \in E$,*

$$f'(\bar{x}; v) = \max_{\phi \in \partial f(\bar{x})} \langle \phi, v \rangle.$$

In particular,

$$\partial f(\bar{x}) \neq \emptyset.$$

Worked Example: Subdifferential of a Maximum

Let $f(x) = \max\{x, -2x + 3\}$. The two affine pieces intersect at $x = 1$. Away from the intersection, the active slope is the unique subgradient: $\partial f(x) = \{-2\}$ for $x < 1$ and $\partial f(x) = \{1\}$ for $x > 1$. At $x = 1$, the max formula gives

$$\partial f(1) = \operatorname{conv}\{-2, 1\} = [-2, 1]. \tag{3.5.1}$$

Because $0 \in \partial f(1)$, the point $x = 1$ is a global minimizer with value one. The active-gradient convex combination is not merely a technical description: choosing weights $(1/3, 2/3)$ on slopes -2 and 1 produces the zero optimality certificate.

3.6 Duality of Points and Subgradients (Fenchel Duality)

Given $\bar{x} \in \operatorname{int}(\operatorname{dom} f)$, ask for $\phi \in \partial f(\bar{x})$.

Reverse question: Given $\phi \in E$, find $\bar{x}$ such that $\phi \in \partial f(\bar{x})$.

Claim:

$$\bar{x} \text{ globally minimizes } f \quad \Longleftrightarrow \quad 0 \in \partial f(\bar{x}).$$

Proof.

$$0 \in \partial f(\bar{x}) \iff \forall x,\ f(x) \geq f(\bar{x}) + \langle 0, x - \bar{x} \rangle = f(\bar{x}) \iff \bar{x} \text{ is a global minimizer.}$$

□

Reverse formulation:

$$\phi \in \partial f(\bar{x}) \iff 0 \in \partial(f - \langle \phi, \cdot \rangle)(\bar{x})$$
$$\iff \bar{x} \text{ minimizes } f(x) - \langle \phi, x \rangle$$
$$\iff \bar{x} \text{ maximizes } \langle \phi, x \rangle - f(x).$$

Fenchel Conjugate

This supremum is called the **Fenchel conjugate**:

$$f^*(\phi) = \sup_{x \in E} \{ \langle \phi, x \rangle - f(x) \}.$$

Properties of f^*

(1) If $\bar{x}$ attains $\sup \langle \phi, x \rangle - f(x)$, then $\phi \in \partial f(\bar{x})$.

(2) If $\phi \in \partial f(\bar{x})$, then $\bar{x}$ attains the supremum.

(3) For all x, ϕ:

$$f^*(\phi) + f(x) \geq \langle \phi, x \rangle.$$

(Fenchel–Young inequality)

$$f^*(\phi) + f(x) \geq \langle \phi, x \rangle,$$

with equality iff $\phi \in \partial f(x)$.

(4) f^* is convex (regardless of f).

(5) f^* is closed.

(6) $f^{**} = f$ iff f is closed and convex.

Examples

(1) Linear function

$$f(x) = \langle c, x \rangle \quad \Rightarrow \quad f^*(\phi) = \begin{cases} 0 & \phi = c, \\ +\infty & \text{otherwise.} \end{cases}$$

(2) Indicator at a point

$$f(x) = \begin{cases} 0 & x = c, \\ +\infty & \text{else,} \end{cases} \quad \Rightarrow \quad f^*(\phi) = \langle c, \phi \rangle.$$

(3) Nonnegative indicator

$$f(x) = \begin{cases} 0 & x \geq 0, \\ +\infty & \text{else}, \end{cases} \quad \Rightarrow \quad f^*(\phi) = \begin{cases} 0 & \phi \leq 0, \\ +\infty & \text{else}. \end{cases}$$

(4) Power function

$$f(x) = \frac{1}{p}|x|^p \quad \Longleftrightarrow \quad f^*(\phi) = \frac{1}{q}|\phi|^q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

(5)

$$f(x) = \sqrt{1+x^2} \quad \Longleftrightarrow \quad f^*(\phi) = \begin{cases} -\sqrt{1-\phi^2} & |\phi| \leq 1, \\ +\infty & \text{else}. \end{cases}$$

Convexity of f^*

$$\lambda f^*(\phi) + (1-\lambda) f^*(\psi) \geq \langle \lambda\phi + (1-\lambda)\psi, x\rangle - f(x)$$
$$\Rightarrow f^*(\lambda\phi + (1-\lambda)\psi) \leq \lambda f^*(\phi) + (1-\lambda) f^*(\psi).$$

Closedness of f^*

$$f^*(\phi_i) \geq \langle \phi_i, x\rangle - f(x)$$
$$\Rightarrow \liminf f^*(\phi_i) \geq \langle \phi, x\rangle - f(x)$$
$$\Rightarrow \liminf f^*(\phi_i) \geq \sup_x(\langle \phi, x\rangle - f(x)) = f^*(\phi).$$

Fenchel Duality

Theorem 3.6.1 (Fenchel Duality)**.** *For $f : E \to (-\infty, \infty]$, $g : Y \to (-\infty, \infty]$, and linear $A : E \to Y$:*

$$p^* = \inf_{x\in E}\{f(x) + g(Ax)\},$$
$$d^* = \sup_{\phi\in Y}\{-f^*(A^T\phi) - g^*(-\phi)\}.$$

In general,

$$p^* \geq d^*.$$

If f, g are convex and

$$0 \in \operatorname{int}(\operatorname{dom} g - A \operatorname{dom} f),$$

then

$$p^* = d^*, \quad d^* \text{ attained.}$$

Proof. By Fenchel–Young:

$$f(x) + f^*(A^T\phi) \geq \langle x, A^T\phi\rangle = \langle Ax, \phi\rangle,$$

$$g(Ax) + g^*(-\phi) \geq \langle Ax, -\phi\rangle.$$

Summing:

$$f(x) + g(Ax) + f^*(A^T\phi) + g^*(-\phi) \geq 0.$$

Thus

$$f(x) + g(Ax) \geq -f^*(A^T\phi) - g^*(-\phi).$$

Taking inf over x and sup over ϕ:

$$p^* \geq d^*.$$

□

Perturbation View

Define

$$h(u) = \inf_x\{f(x) + g(Ax + u)\}, \quad h(0) = p^*.$$

Then h is convex and

$$\operatorname{dom} h = \operatorname{dom} g - A \operatorname{dom} f.$$

There exists $-\phi \in \partial h(0)$, so

$$h(0) \leq h(u) + \langle \phi, u\rangle.$$

$$\Rightarrow h(0) \leq f(x) + g(Ax + u) + \langle \phi, u\rangle$$

$$= f(x) - \langle A^T\phi, x\rangle + g(Ax + u) - \langle \phi, Ax + u\rangle.$$

Taking inf:

$$h(0) \leq -f^*(A^T\phi) - g^*(-\phi) = d^* \leq p^*.$$

Worked Example: Fenchel Dual of the Lasso

Consider

$$\min_{x\in\mathbb{R}^n} \frac{1}{2}\|Ax - b\|_2^2 + \lambda\|x\|_1. \tag{3.6.1}$$

Introduce $r = Ax - b$ and a multiplier y for $r - Ax + b = 0$. The Lagrangian is

$$\frac{1}{2}\|r\|^2 + \lambda\|x\|_1 + y^T(r - Ax + b). \tag{3.6.2}$$

Minimizing in r gives $r = -y$ and contribution $-\frac{1}{2}\|y\|^2$. Minimizing in x is finite precisely when $\left\|A^T y\right\|_\infty \le \lambda$, because $(\lambda\|\cdot\|_1)^* = \iota_{\{u:\|u\|_\infty \le \lambda\}}$. Thus the dual is

$$\max_y \; b^T y - \frac{1}{2}\|y\|_2^2 \quad \text{subject to} \quad \left\|A^T y\right\|_\infty \le \lambda, \tag{3.6.3}$$

after replacing y by $-y$ if the opposite multiplier convention is used. At optimality, $y^\star = b - Ax^\star$ and

$$A^T y^\star \in \lambda \partial \|x^\star\|_1 . \tag{3.6.4}$$

Hence $|(A^T y^\star)_j| < \lambda$ forces $x_j^\star = 0$, while a nonzero coefficient has correlation exactly λ with the residual, with the appropriate sign.

3.7 Biconjugates and Lagrange Duality

3.7.1 Stationarity via Subgradients

Recall

$$f^*(\phi) = \sup_{x \in E}\{\langle \phi, x\rangle - f(x)\}.$$

We want:

$$0 \in \partial\Big(f + g \circ A\Big)(\bar{x}).$$

Equivalently,

$$\exists\, \psi \in \partial f(\bar{x}), \quad \phi \in \partial g(A\bar{x}) \quad \text{s.t.} \quad \psi + A^T\phi = 0.$$

Recall the dual:

$$d^* = \sup_\phi \Big(- f^*(A^T\phi) - g^*(-\phi)\Big).$$

3.7.2 Fenchel–Young Symmetry

Suppose $f = f^{**}$.

Then

$$f^*(\phi) + f(\bar{x}) = \langle \phi, \bar{x}\rangle \quad \Longleftrightarrow \quad \phi \in \partial f(\bar{x}).$$

Also

$$f^{**}(\bar{x}) + f^*(\phi) = \langle \bar{x}, \phi\rangle \quad \Longleftrightarrow \quad \bar{x} \in \partial f^*(\phi).$$

If f, f^* are C^1, then

$$\phi = \nabla f(x) \quad \Longleftrightarrow \quad x = \nabla f^*(\phi),$$

so

$$\nabla f^* = (\nabla f)^{-1}.$$

3.7.3 Biconjugate Inequality

Lemma 3.7.1. *For any $h : E \to (-\infty, \infty]$,*

$$h^{**}(x) \le h(x), \quad \forall x.$$

Proof. By Fenchel–Young:

$$\langle x, \phi \rangle - h^*(\phi) \le h(x).$$

Taking supremum over ϕ:

$$h^{**}(x) = \sup_{\phi} \{\langle x, \phi \rangle - h^*(\phi)\} \le h(x).$$

□

3.7.4 Biconjugate as Convex Envelope

The biconjugate h^{**} is the largest closed convex lower bound of h.

$$\operatorname{epi} h^{**} = \operatorname{cl}\Big(\operatorname{conv}(\operatorname{epi} h)\Big).$$

3.7.5 Affine Minorants

Let

$$\alpha(x) = \langle \phi, x \rangle + c$$

be an affine minorant of h if $\alpha(x) \le h(x)$.

Theorem 3.7.1. *For any $h : E \to (-\infty, \infty]$, the following are equivalent:*

(i) *h is closed and convex*

(ii) *$h = h^{**}$*

(iii) *$h(x) = \sup\{\alpha(x) \mid \alpha \text{ affine minorant of } h\}$*

3.7.6 Lagrange Duality via Value Function

Recall the Lagrangian:

$$L(x, \lambda) = f(x) + \sum_i \lambda_i g_i(x).$$

$$p^* = \inf_x \sup_{\lambda \ge 0} L(x, \lambda), \quad d^* = \sup_{\lambda \ge 0} \inf_x L(x, \lambda).$$

Define the value function:

$$v(\Delta) = \inf_x \{f(x) \mid g_i(x) \le \Delta_i\}.$$

Then:

(1) $p^* = v(0)$

(2) $v^*(-\lambda) = \begin{cases} -\Phi(\lambda) & \lambda \geq 0 \\ +\infty & \text{otherwise} \end{cases}$

(3) $d^* = v^{**}(0)$

Thus weak duality follows from:

$$v^{**} \leq v.$$

3.7.7 Strong Duality via Regularity

If f, g_i are convex, then v is convex.

We need:

$$\liminf_{\Delta \to 0} v(\Delta) \geq v(0)$$

(lower semicontinuity at 0).

This is implied by existence of a strictly feasible point:

$$\exists\ x_s \text{ such that } g_i(x_s) < 0 \quad \forall i.$$

Then:

$$0 \in \text{int}(\text{dom}\, v) \Rightarrow \exists \phi \in \partial v(0) \Rightarrow v \text{ is l.s.c. at } 0 \Rightarrow v^{**}(0) = v(0).$$

Hence:

$$p^* = d^*.$$

Worked Example: Computing a Biconjugate

Let f be zero at $x = -1$ and $x = 1$ and $+\infty$ elsewhere. Its conjugate is

$$f^*(y) = \max\{-y, y\} = |y|. \tag{3.7.1}$$

Therefore

$$f^{**}(x) = \sup_y \{xy - |y|\} = \begin{cases} 0, & |x| \leq 1, \\ +\infty, & |x| > 1. \end{cases} \tag{3.7.2}$$

The biconjugate fills in the entire segment between the two finite points. It is the closed convex envelope of f, not a reconstruction of the original nonconvex effective domain.

Chapter 4

Methods

4.1 Subgradient and Bundle Methods

Basic Setup

Problem:

$$\min_x f(x), \quad f : E \to \mathbb{R} \text{ convex.}$$

Algorithm idea: At each iteration compute

$$g_k \in \partial f(x_k).$$

(Note: not our choice.)

Example

$$f(x) = \max_i \{\langle a_i, x\rangle + b_i\} + \tfrac{1}{2}\|x\|^2.$$

Then

$$\partial f(x) \ni a_i + x, \quad \text{for any } i \text{ attaining the maximum.}$$

4.1.1 Subgradient Method

$$x_{k+1} = x_k - \alpha_k g_k, \quad g_k \in \partial f(x_k).$$

For $f \notin C^1$, it may happen that

$$f(x_{k+1}) > f(x_k) \quad \text{for all } \alpha_k > 0.$$

Proximal View

$$x_{k+1} = \arg\min_y \left\{ f(y) + \frac{1}{2\alpha_k}\|y - x_k\|^2 \right\}.$$

Linearization:

$$f(y) \approx f(x_k) + \langle g_k, y - x_k \rangle.$$

Then

$$x_{k+1} = \arg\min_y \left\{ f(x_k) + \langle g_k, y - x_k \rangle + \frac{1}{2\alpha_k}\|y - x_k\|^2 \right\}.$$

Optimality:

$$g_k + \frac{1}{\alpha_k}(y - x_k) = 0 \quad \Longrightarrow \quad y = x_k - \alpha_k g_k.$$

Interpretation:

$$0 \in \partial f(x_k) \quad \text{(approximately)}.$$

4.1.2 Dual Averaging

Define

$$\ell_k(x) = \frac{\sum_{i=1}^k \lambda_i \big(f(x_i) + \langle g_i, x - x_i \rangle \big)}{\sum_{i=1}^k \lambda_i}, \quad g_i \in \partial f(x_i).$$

Update:

$$x_{k+1} = \arg\min_y \left\{ \ell_k(y) + \frac{\beta_k}{2\sum \lambda_i}\|y - x_0\|^2 \right\}.$$

This gives:

$$x_{k+1} = x_0 - \frac{\sum \lambda_i g_i}{\beta_k} = x_0 - \sum \alpha_i g_i.$$

Stopping Criterion

$$f(x_k) - \ell_k(x_k) \le \varepsilon, \quad \|\nabla \ell_k(x_k)\| \le \varepsilon.$$

Constrained Case

$$\min f(x) \quad \text{s.t. } x \in S.$$

Projected subgradient method:

$$x_{k+1} = \arg\min_{y \in S} \left\{ f(x_k) + \langle g_k, y - x_k \rangle + \frac{1}{2\alpha_k}\|y - x_k\|^2 \right\}$$

$$= \text{proj}_S(x_k - \alpha_k g_k).$$

4.1.3 Bundle Methods

Build model m_k:

$$z_{k+1} = \arg\min_y \left\{ m_k(y) + \frac{1}{2\alpha_k}\|y - x_k\|^2 \right\}.$$

Descent step:

$$f(z_{k+1}) \leq f(x_k) - \text{margin} \Rightarrow x_{k+1} = z_{k+1}.$$

Null step:

$$x_{k+1} = x_k,$$

$$m_{k+1}(y) = \max\left\{ m_k(y), f(z_{k+1}) + \langle g_{k+1}, y - z_{k+1} \rangle \right\}, \quad g_{k+1} \in \partial f(z_{k+1}).$$

Steepest Descent Direction

$$\min_{\|v\|\leq 1} f'(x_k; v) = \min_{\|v\|\leq 1} \max_{g \in \partial f(x_k)} \langle g, v \rangle.$$

By strong duality:

$$= \max_{g \in \partial f(x_k)} \min_{\|v\|\leq 1} \langle g, v \rangle.$$

Inner minimization:

$$\min_{\|v\|\leq 1} \langle g, v \rangle = -\|g\|, \quad \text{attained at } v = -\frac{g}{\|g\|}.$$

Thus:

$$= - \min_{g \in \partial f(x_k)} \|g\|.$$

4.1.4 Subgradient Method for Constraints

If x_k feasible:

$$x_{k+1} = x_k - \alpha_k \phi_k, \quad \phi_k \in \partial f(x_k).$$

If infeasible:

$$x_{k+1} = x_k - \alpha_k \phi_k, \quad \phi_k \in \partial g_i(x_k), \quad g_i(x_k) > 0.$$

Goal:

$$f_{\text{best}}^{(k)} = \min_{j \leq k,\ x_j \text{ feasible}} f(x_j) \to f(x^*).$$

Step Size Assumptions

$$\alpha_k > 0, \quad \sum \alpha_k = \infty, \quad \sum \alpha_k^2 < \infty, \quad \alpha_k = \frac{1}{k}.$$

Assume:

- f, g_i convex
- bounded subgradients $\|g_k\| \leq G$
- Slater point $\tilde{x}$ with $g_i(\tilde{x}) < 0$

Let

$$R \geq \|x_0 - x^*\|, \ \|x_0 - \tilde{x}\|.$$

Convergence Result

$$f_{\text{best}}^{(k)} \to f(x^*).$$

Key Inequality

For feasible x_k:

$$\phi_k^T(x_k - \tilde{x}) \geq f(x_k) - f(\tilde{x}).$$

For infeasible:

$$\phi_k^T(x_k - \tilde{x}) \geq g_i(x_k) - g_i(\tilde{x}) \geq \mu > 0.$$

Thus:

$$\|x_{k+1} - \tilde{x}\|^2 \leq \|x_k - \tilde{x}\|^2 - \delta\alpha_k + G^2\alpha_k^2.$$

Summing:

$$0 \leq R^2 - \delta \sum \alpha_k + G^2 \sum \alpha_k^2.$$

$$\Rightarrow \delta \leq \frac{R^2 + G^2 \sum \alpha_k^2}{\sum \alpha_k} \to 0.$$

Contradiction $\Rightarrow$ convergence.

Lagrangian Interpretation

$$\min_x \max_{\lambda \geq 0} \left\{ f(x) + \sum \lambda_i g_i(x) \right\}.$$

Stopping when:

$$\|\nabla f + \sum \lambda_i \nabla g_i\| \leq \varepsilon.$$

Worked Example: Subgradient Iteration for an Absolute Value

For $f(x) = |x - 2|$ on $[-1, 4]$, take $x_0 = -1$ and $g_k \in \partial f(x_k)$. With constant step $\alpha = 3/4$, projected subgradient descent gives

$$x_{k+1} = \Pi_{[-1,4]}(x_k - \alpha g_k). \tag{4.1.1}$$

As long as $x_k < 2$, $g_k = -1$, so the first iterates are $x_1 = -1/4$, $x_2 = 1/2$, $x_3 = 5/4$, and $x_4 = 2$. If an iterate crosses the kink, the subgradient changes sign and a constant step can oscillate in an $O(\alpha)$ neighborhood. A diminishing nonsummable step eliminates this fixed error floor. The calculation is elementary, but it displays why the subgradient method generally controls best objective values or averages rather than producing monotone descent.

4.2 Penalty and Lagrangian Methods

4.2.1 Penalty Method

$$P_* \approx \min_x \left\{ f(x) + \rho \sum_{i=1}^{m} \max(0, g_i(x))^2 \right\}.$$

If $f, g_i \in C^1$, then

$$f(x) + \rho \sum_i \max(0, g_i(x))^2 \in C^1.$$

Let

$$\bar{x}(\rho) \in \arg\min \left\{ f(x) + \rho \sum_i \max(0, g_i(x))^2 \right\}.$$

Theorem 4.2.1. *If $\bar{x}(\rho) \to x_*$ as $\rho \to \infty$, then x_* globally minimizes f over $\{g_i(x) \le 0\}$.*

Proof.

$$f(\bar{x}(\rho)) + \sum \rho \max(0, g_i(\bar{x}(\rho)))^2 \le P_*, \quad \Rightarrow \quad f(\bar{x}(\rho)) \le P_*.$$

If for contradiction $g_i(\bar{x}(\rho)) \ge \varepsilon > 0$, then

$$f(\bar{x}(\rho)) + \rho\varepsilon^2 \to \infty,$$

a contradiction.

Thus $\bar{x}(\rho)$ must approach feasibility, and the result follows. □

Exact Penalization

$$\operatorname{dist}_C(x) = \min_{y \in C} \|y - x\|.$$

$$\min f(x) \quad \text{s.t. } x \in C.$$

Assume:

- f is globally M-Lipschitz
- dist_C is tractable

Solve:

$$\min \{f(x) + M' \text{dist}_C(x)\}, \quad M' > M.$$

Theorem 4.2.2. *This problem admits minimizers with $x \in C$.*

Proof. Let $x \notin C$ and $y_i \to C$ with $\|y_i - x\| \to \text{dist}_C(x)$.

$$f(y_i) \leq f(x) + M\|x - y_i\| < f(x) + M'\|x - y_i\|.$$

Taking limits:

$$\lim f(y_i) \leq f(x) + M' \text{dist}_C(x).$$

Since $\text{dist}_C(y_i) = 0$, feasible points dominate. □

4.2.2 Augmented Lagrangian Method

$$\min f(x) \quad \text{s.t. } Ax = b.$$

$$\mathcal{L}_\rho(x, \lambda) = f(x) + \lambda^T(Ax - b) + \frac{\rho}{2}\|Ax - b\|^2.$$

$$\min_x \max_\lambda \mathcal{L}_\rho(x, \lambda).$$

Primal-Dual Iteration

$$z_k = \begin{bmatrix} x_k \\ \lambda_k \end{bmatrix}, \quad z_{k+1} = z_k - \alpha_k T(z_k),$$

$$T(z_k) \in \begin{bmatrix} \partial f(x_k) + A^T\lambda_k + \rho A^T(Ax_k - b) \\ b - Ax_k \end{bmatrix}.$$

Convergence

Theorem 4.2.3. *Suppose f is convex, $\|\phi_k\| \leq G$, and iterates are bounded. If*

$$\sum \gamma_k = \infty, \quad \sum \gamma_k^2 < \infty,$$

with $\alpha_k = \gamma_k / \|T(z_k)\|$, then

$$f(x_k) \to f(x_*), \quad \|Ax_k - b\| \to 0.$$

Proof.

$$\|z_{k+1} - z_*\|^2 = \|z_k - z_*\|^2 - 2\gamma_k \frac{T(z_k)^T(z_k - z_*)}{\|T(z_k)\|} + \gamma_k^2.$$

Summing yields boundedness and convergence once we show:

$$T(z_k)^T(z_k - z_*) \geq 0.$$

Compute:

$$T(z_k)^T(z_k - z_*) = \phi_k^T(x_k - x_*) + \lambda_k^T(Ax_k - b) + \rho\|Ax_k - b\|^2 - (Ax_k - b)^T(\lambda_k - \lambda_*).$$

$$= \phi_k^T(x_k - x_*) + \lambda_*^T(Ax_k - b) + \rho\|Ax_k - b\|^2$$

$$\geq f(x_k) - f(x_*) + \lambda_*^T(Ax_k - b) + \rho\|Ax_k - b\|^2 \geq 0.$$

Thus the claim holds and convergence follows. □

Worked Example: Exact Penalty Threshold

Consider

$$\min_x (x-3)^2 \qquad \text{subject to} \qquad x \leq 1. \tag{4.2.1}$$

The constrained solution is $x^\star = 1$. Stationarity for $L(x, \lambda) = (x-3)^2 + \lambda(x-1)$ gives $2(x^\star - 3) + \lambda^\star = 0$, hence $\lambda^\star = 4$. The ℓ_1 penalty is

$$P_\rho(x) = (x-3)^2 + \rho(x-1)_+. \tag{4.2.2}$$

For $x > 1$, its stationary point is $x = 3 - \rho/2$. If $\rho < 4$, this point is infeasible and has lower penalized value than the boundary. If $\rho \geq 4$, the right derivative at 1 is $-4 + \rho \geq 0$, while the left derivative is $-4 < 0$, so $x = 1$ minimizes P_ρ. The smallest exact penalty parameter is therefore the optimal multiplier magnitude. Squared penalties do not share this finite-threshold property.

4.3 Decomposition Methods

4.3.1 Primal Decomposition

Consider

$$\min_{x_1, x_2, y} f_1(x_1, y) + f_2(x_2, y).$$

Rewrite:

$$= \min_y \left(\inf_{x_1} f_1(x_1, y) \right) + \left(\inf_{x_2} f_2(x_2, y) \right).$$

Exercise 4.3.1. If f_i is convex, then

$$\phi_i(y) = \inf_{x_i} f_i(x_i, y)$$

is convex.

Primal Decomposition Method

Solve in parallel:

- Find x_1 minimizing ϕ_1-problem and $g_1 \in \partial\phi_1(y_k)$
- Find x_2 minimizing ϕ_2-problem and $g_2 \in \partial\phi_2(y_k)$

Update:

$$y_{k+1} = y_k - \alpha_k(g_1 + g_2).$$

At runtime:

$$\text{OPT} \leq f_1(x_1^*, y_k) + f_2(x_2^*, y_k).$$

4.3.2 Dual Decomposition

$$\min_{x_1,x_2,y} f_1(x_1, y) + f_2(x_2, y)$$

$$= \min_{x_1,x_2,y_1,y_2} f_1(x_1, y_1) + f_2(x_2, y_2) \quad \text{s.t. } y_1 = y_2.$$

Lagrangian:

$$= \min_{x_i,y_i} \max_{\lambda} f_1(x_1, y_1) + f_2(x_2, y_2) + \lambda^T(y_1 - y_2).$$

Dual Function

$$\Phi(\lambda) = \underbrace{\min_{x_1,y_1} \left(f_1(x_1, y_1) + \lambda^T y_1\right)}_{\Phi_1(\lambda)} + \underbrace{\min_{x_2,y_2} \left(f_2(x_2, y_2) - \lambda^T y_2\right)}_{\Phi_2(\lambda)}.$$

$$\Phi_1(\lambda) = -(f_1)^*(0, -\lambda), \quad \Phi_2(\lambda) = -(f_2)^*(0, \lambda).$$

If x_i, y_i minimize the subproblems:

$$-y_1 \in \partial\Phi_1(\lambda), \quad y_2 \in \partial\Phi_2(\lambda),$$

$$\Rightarrow y_2 - y_1 \in \partial(-\Phi_1 - \Phi_2)(\lambda).$$

Dual Decomposition Method

Solve in parallel:

$$x_1, y_1 = \arg\min f_1(x_1, y_1) + \lambda_k^T y_1,$$
$$x_2, y_2 = \arg\min f_2(x_2, y_2) - \lambda_k^T y_2.$$

Update:

$$\lambda_{k+1} = \lambda_k - \alpha_k(y_2 - y_1), \quad (y_2 - y_1) \in \partial\Phi(\lambda_k).$$

$$\text{OPT} \geq \text{Dual OPT} \geq \Phi_1(\lambda_k) + \Phi_2(\lambda_k).$$

4.3.3 ADMM (Alternating Direction Method of Multipliers)

$$\min f(x) + g(z) \quad \text{s.t. } Ax + Bz = c.$$

Augmented Lagrangian:

$$\mathcal{L}_\rho(x, y, z) = f(x) + g(z) + y^T(Ax + Bz - c) + \frac{\rho}{2}\|Ax + Bz - c\|^2.$$

Updates:

$$x_{k+1} \in \arg\min_x \mathcal{L}_\rho(x, y_k, z_k),$$
$$z_{k+1} \in \arg\min_z \mathcal{L}_\rho(x_{k+1}, y_k, z),$$
$$y_{k+1} = y_k + \rho(Ax_{k+1} + Bz_{k+1} - c).$$

4.3.4 Applications

4.3.4.1 Consensus Optimization

$$\min_x \sum_i f_i(x) = \min_{x_i, z} \sum_i f_i(x_i) \quad \text{s.t. } x_i = z.$$

ADMM updates:

$$x_i^{k+1} \in \arg\min f_i(x_i) + y_i^{kT}(x_i - z_k) + \frac{\rho}{2}\|x_i - z_k\|^2,$$

$$z^{k+1} = \frac{1}{m}\sum_i \left(x_i^{k+1} + \frac{1}{\rho}y_i^k\right),$$

$$y_i^{k+1} = y_i^k + \rho(x_i^{k+1} - z^{k+1}).$$

4.3.4.2 Exchange Optimization

$$\min \sum_i f_i(x_i) \quad \text{s.t.} \ \sum_i x_i = 0.$$

4.3.4.3 Example: Power System

Each x_i is production at facility i:

$$f_i(x_i) = \text{cost.}$$

$$\min \sum_i f_i(x_i) \quad \text{s.t.} \ \sum_i x_i = \text{Target.}$$

$$= \min_x \max_y \sum_i f_i(x_i) + y \left(\sum_i x_i - \text{Target} \right).$$

4.3.5 Network Flow Example

Directed graph with p nodes and n arcs.

Let x_j be flow on arc j:

$$x_j > 0 \text{ forward}, \quad x_j < 0 \text{ reverse}.$$

At node i:

$$s_i > 0 \text{ inflow}, \quad s_i < 0 \text{ outflow}.$$

Flow conservation:

$$Ax + s = 0,$$

where A is the incidence matrix.

Costs

Each arc has cost $\phi_j(x_j)$ (strictly convex).

Example (M/M/1 queue):

$$\phi_j(x_j) = \begin{cases} \frac{x_j}{c_j - x_j} & x_j \in [0, c_j) \\ +\infty & \text{otherwise.} \end{cases}$$

Optimization Problem

$$\min \sum_j \phi_j(x_j) \quad \text{s.t.} \ Ax + s = 0.$$

Dual Decomposition

$$\mathcal{L}(x, v) = \sum_j \phi_j(x_j) + v^T(Ax + s).$$

$$= v^T s + \sum_j \left(\phi_j(x_j) + (A_j^T v)x_j\right).$$

Let

$$(\Delta v)_j = A_j^T v.$$

Dual function:

$$\Phi(v) = v^T s + \sum_j \inf_{x_j} \left(\phi_j(x_j) + (\Delta v)_j x_j\right).$$

$$= v^T s - \sum_j \sup_{x_j} \left(-(\Delta v)_j x_j - \phi_j(x_j)\right).$$

$$= v^T s - \sum_j \phi_j^*(-(\Delta v)_j).$$

Dual Problem

$$\max_v \left\{ v^T s - \sum_j \phi_j^*(-(\Delta v)_j) \right\}.$$

If ϕ_j strictly convex, then ϕ_j^* has unique maximizer.

Let $x_j^*(v)$ denote it.

Claim.

$$-(Ax^*(\Delta v) + s) \in \partial(-\Phi)(v).$$

4.3.6 Special Case (LP)

$$\min f(x) + g(z) \quad \text{s.t. } Ax = z,$$

$$f(x) = \begin{cases} \langle c, x\rangle & x \geq 0 \\ +\infty & \text{else,} \end{cases} \qquad g(z) = \begin{cases} 0 & z = b \\ +\infty & \text{else.} \end{cases}$$

$$\iff \min \langle c, x\rangle \quad \text{s.t. } Ax = b,\ x \geq 0.$$

Worked Example: Consensus ADMM by Hand

Minimize

$$\frac{1}{2}(x-1)^2+\frac{1}{2}(z+1)^2 \quad \text{subject to} \quad x=z. \tag{4.3.1}$$

The solution is $x^\star = z^\star = 0$. With scaled multiplier u, penalty $\rho = 1$, and initial $(z_0, u_0) = (0, 0)$, ADMM gives

$$x_{k+1} = \arg\min_x \frac{1}{2}(x-1)^2+\frac{1}{2}(x-z_k+u_k)^2 = \frac{1+z_k-u_k}{2}, \tag{4.3.2}$$

$$z_{k+1} = \arg\min_z \frac{1}{2}(z+1)^2+\frac{1}{2}(x_{k+1}-z+u_k)^2 = \frac{x_{k+1}+u_k-1}{2}, \tag{4.3.3}$$

$$u_{k+1} = u_k + x_{k+1} - z_{k+1}. \tag{4.3.4}$$

The first update is $(x_1, z_1, u_1) = (1/2, -1/4, 3/4)$; the second is $(x_2, z_2, u_2) = (0, -1/8, 7/8)$. The primal residual $x_k - z_k$ and dual residual $\rho(z_k - z_{k-1})$ measure different failures and should both be checked. The iterates need not decrease the original objective monotonically.

Worked Example: Dual Decomposition for Resource Allocation

Two agents solve

$$\min_{x_1,x_2\geq 0} \frac{1}{2}(x_1-2)^2+\frac{1}{2}(x_2-4)^2 \quad \text{subject to} \quad x_1+x_2 \leq 3. \tag{4.3.5}$$

With multiplier $\lambda \geq 0$, the agent subproblems separate and give

$$x_1(\lambda) = (2-\lambda)_+, \qquad x_2(\lambda) = (4-\lambda)_+. \tag{4.3.6}$$

The dual subgradient is $x_1(\lambda) + x_2(\lambda) - 3$. Since the optimum has both variables positive, solving $6 - 2\lambda = 3$ gives $\lambda^\star = 3/2$ and $(x_1^\star, x_2^\star) = (1/2, 5/2)$. The common price coordinates otherwise independent agents and enforces the shared capacity.

4.4 Interior Point and Newton Methods

4.4.1 Barrier Method (LP Sketch)

Consider the linear program:

$$P^\star = \min c^T x \quad \text{s.t. } Ax = b,\ x \geq 0.$$

Introduce barrier:

$$f(x) = \begin{cases} -\sum_{i=1}^n \log(x_i) & x > 0 \\ +\infty & \text{else.} \end{cases}$$

Solve:

$$x_\eta^\star \in \arg\min\ \eta\, c^T x + f(x) \quad \text{s.t. } Ax = b, \quad \eta > 0.$$

Goal:

$$x_\eta^\star \to x^\star \quad \text{as } \eta \to \infty.$$

First-Order Optimality

$$\nabla(\eta c^T x + f(x))(x_\eta^\star) \perp \{x : Ax = b\}$$

$$\Rightarrow \nabla(\eta c^T x + f(x))(x_\eta^\star) = A^T y_\eta^\star.$$

Theorem 4.4.1.

$$b^T y_\eta^\star \geq c^T x_\eta^\star - \frac{n}{\eta}.$$

Thus large η gives near-optimal solutions.

Central Path

As η increases, $x_\eta^\star$ traces the central path.

Barrier Method Algorithm

Given $\eta_0 > 0$ and x_0 approximately solving:

$$\min \eta_0 c^T x + f(x) \quad \text{s.t. } Ax = b,$$

iterate:

$$\eta_{k+1} = \alpha \eta_k, \quad \alpha > 1,$$

$$x_{k+1} \approx \arg\min \eta_{k+1} c^T x + f(x) \quad \text{s.t. } Ax = b,$$

using warm-start from x_k.

Each stage uses a few Newton steps.

Differentiability and Geometry

We work in Euclidean space E with inner product:

$$\langle x, y \rangle = x^T y.$$

For matrices:

$$\langle X, Y \rangle = \operatorname{tr}(X^T Y).$$

Definition 4.4.1. f is (Fréchet) differentiable at x if

$$\lim_{\Delta \to 0} \frac{|f(x+\Delta) - (f(x) + \langle g, \Delta \rangle)|}{\|\Delta\|} = 0.$$

Example: Log-Det Barrier

$$f(X) = \begin{cases} -\ln\det(X) & X \succ 0 \\ +\infty & \text{else.} \end{cases}$$

Theorem 4.4.2.

$$\nabla f(X) = -X^{-1}.$$

Proof. Using eigenvalues λ_i of $X^{-1}\Delta$:

$$f(X+\Delta) - f(X) = -\sum_i \ln(1+\lambda_i),$$

and

$$\langle -X^{-1}, \Delta\rangle = -\sum_i \lambda_i.$$

Using $\ln(1+\lambda) \approx \lambda$ gives result. □

Inner Product Dependence

Theorem 4.4.3. *If f is differentiable under $\langle\cdot,\cdot\rangle$ with gradient g, then under*

$$\langle x, y\rangle_S = \langle x, Sy\rangle,$$

the gradient is $S^{-1}g$.

Proof.

$$f(x+\Delta) - f(x) \approx \langle g, \Delta\rangle = \langle S^{-1}g, \Delta\rangle_S.$$

□

Thus:

Differentiability is inner-product invariant, but gradients are not.

Hessian

Definition 4.4.2. The Hessian $H(x)$ satisfies:

$$\lim_{\Delta\to 0} \frac{\|g(x+\Delta) - (g(x) + H(x)\Delta)\|}{\|\Delta\|} = 0.$$

Example

$$f(X) = -\ln\det(X) \Rightarrow \nabla^2 f(X)[\Delta] = X^{-1}\Delta X^{-1}.$$

4.4.2 Newton's Method

Theorem 4.4.4. *If z minimizes f, then the Newton step $x^+ = x + n(x)$ satisfies:*

$$\|x^+ - z\| \leq \|H(x)^{-1}\| \int_0^1 \|H(x + t(z - x)) - H(x)\| dt \cdot \|x - z\|.$$

Proof. Use $g(z) = 0$ and the fundamental theorem:

$$g(x) - g(z) = \int_0^1 H(x + t(z - x))(z - x)\, dt.$$

Substitute into Newton update. □

Theorem 4.4.5. *If H is Q-Lipschitz, then:*

$$\|x^+ - z\| \leq \|H(x)^{-1}\| \frac{Q}{2} \|x - z\|^2.$$

Thus Newton has quadratic convergence.

4.4.3 Newton with Constraints

Let

$$L = \{x : Ax = b\}.$$

Optimality:

$$-\nabla f(x) \in N_L(x) \quad \Longleftrightarrow \quad -\nabla f(x) = A^T y.$$

Linearization:

$$-\nabla f(x + \Delta) \approx -(\nabla f(x) + \nabla^2 f(x)\Delta) = A^T y.$$

Solve:

$$\min f(x) + \langle g(x), \Delta \rangle + \frac{1}{2} \langle \Delta, H(x)\Delta \rangle \quad \text{s.t. } A(x + \Delta) = b.$$

Projected Form

Let projection onto L:

$$g_L(x) = P_L g(x), \quad H_L(x) = P_L H(x).$$

Theorem 4.4.6. *Newton direction:*

$$n_L(x) = -H_L(x)^{-1} g_L(x).$$

Intrinsic Geometry

Definition 4.4.3. Define intrinsic inner product:

$$\langle y, z\rangle_x = \langle y, H(x)z\rangle.$$

Intrinsic norm:

$$\|y\|_x = \sqrt{\langle y, H(x)y\rangle}.$$

Then:

$$g_x(x) = H(x)^{-1}g(x) = -n(x), \quad H_x(x) = I.$$

4.4.4 Self-Concordance

Definition 4.4.4. Let

$$B_x(y, r) = \{z : \|z - y\|_x \leq r\}.$$

f is self-concordant if:

$$B_x(x, 1) \subset \operatorname{dom} f$$

and for $y \in B_x(x, 1)$:

$$1 - \|y - x\|_x \leq \frac{\|v\|_y}{\|v\|_x} \leq \frac{1}{1 - \|y - x\|_x}.$$

Example

$$f(x) = -\sum_i \log x_i.$$

$$g(x) = \begin{bmatrix} 1/x_1 \\ \vdots \\ 1/x_n \end{bmatrix}, \quad H(x) = \begin{bmatrix} 1/x_1^2 & & \\ & \ddots & \\ & & 1/x_n^2 \end{bmatrix}.$$

For $y \in B_x(x, 1)$:

$$\sum_i \frac{(y_i - x_i)^2}{x_i^2} < 1 \Rightarrow y_i > 0.$$

Also:

$$\|v\|_y^2 \leq \|v\|_x^2 \cdot \max_i \frac{x_i^2}{y_i^2} \leq \frac{1}{(1 - \|y - x\|_x)^2}.$$

Self-Concordance and Newton Behavior

Recall the intrinsic geometry:

$$\langle y, z\rangle_x = \langle y, H(x)z\rangle, \quad \|y\|_x = \sqrt{\langle y, y\rangle_x}.$$

Then:

$$g_x(x) = H(x)^{-1}g(x) = -n(x), \quad H_x(x) = I.$$

Definition 4.4.5. f is self-concordant (SC) if for all $x \in domf$:

1. $B_x(x, 1) \subset domf$
2. For all $y \in B_x(x, 1)$:

$$1 - \|y - x\|_x \leq \frac{\|v\|_y}{\|v\|_x} \leq \frac{1}{1 - \|y - x\|_x} \quad \forall v \neq 0.$$

Theorem 4.4.7. *f is SC iff for all $y \in B_x(x, 1)$:*

$$\|H_x(y)\|_x,\ \|H_x(y)^{-1}\|_x \leq \frac{1}{(1 - \|y - x\|_x)^2}.$$

Equivalently:

$$\|H_x(x) - H_x(y)\|_x \leq \frac{1}{(1 - \|y - x\|_x)^2} - 1.$$

Theorem 4.4.8. *If $f \in$ SC and $y \in B_x(x, 1)$, then the second-order model satisfies:*

$$|f(y) - q_x(y)| \leq \frac{\|y - x\|_x^3}{3(1 - \|y - x\|_x)}.$$

Proof.

$$f(y) - q_x(y) = \int_0^1 \int_0^t \langle (H_x(x + s(y - x)) - I)(y - x), y - x\rangle_x ds dt.$$

Use SC bounds:

$$\|H_x(x + s(y - x)) - I\|_x \leq \frac{1}{(1 - s\|y - x\|_x)^2} - 1.$$

Integrating gives the result. □

Newton Step Contraction

Let:

$$x^+ = x + n(x).$$

Theorem 4.4.9. *If $f \in \mathrm{SC}$ and z is a minimizer with $z \in B_x(x, 1)$, then:*

$$\|x^+ - z\|_x \leq \frac{\|x - z\|_x^2}{1 - \|x - z\|_x}.$$

Proof. Apply previous Newton bound and SC control on Hessian variation. □

Theorem 4.4.10. *If $f \in \mathrm{SC}$ and $\|n(x)\|_x < 1$, then:*

$$\|n(x^+)\|_{x^+} \leq \left(\frac{\|n(x)\|_x}{1 - \|n(x)\|_x}\right)^2.$$

Proof. Use:

$$\|n(x^+)\|_{x^+}^2 = \langle g_x(x^+), H_x(x^+)^{-1} g_x(x^+)\rangle_x,$$

and bound via SC Hessian variation and integral representation of gradients. □

Theorem 4.4.11. *If $f \in \mathrm{SC}$ and*

$$\|n(x)\|_x \leq \frac{1}{4},$$

then f has a minimizer $z \in B_x(x, 1)$ and:

$$\|z - x^+\|_x \leq \frac{3\|n(x)\|_x^2}{(1 - \|n(x)\|_x)^3}.$$

Proof. Using model accuracy:

$$f(y) \geq f(x) - \|n(x)\|_x \|y - x\|_x + \frac{1}{2}\|y - x\|_x^2 - \frac{1}{2}\|y - x\|_x^3.$$

For $\|y - x\|_x \leq \frac{1}{3}$:

$$f(y) \geq f(x),$$

so a minimizer exists in the ball. □

4.4.5 Self-Concordant Barriers

Definition 4.4.6. f is a self-concordant barrier (SCB) if:

$$f \in \mathrm{SC} \quad \text{and} \quad \nu_f := \sup_{x \in dom f} \|g_x(x)\|_x^2 < \infty.$$

Example

$$f(x) = -\sum_i \log x_i \quad \Rightarrow \quad \nu_f = n.$$

Closure Properties

Lemma 4.4.1. *If $f_1, f_2 \in$ SCB, then:*

$$f_1 + f_2 \in \text{SCB}, \quad \nu_{f_1+f_2} \leq \nu_{f_1} + \nu_{f_2}.$$

Lemma 4.4.2. *If $f \in$ SCB, then:*

$$f(Ax - b) \in \text{SCB},$$

with same or smaller ν.

4.4.5.1 Barrier Constructions

Polytope

$$\min c^T x \quad \text{s.t. } Ax \leq b, \ \|x\|_2 \leq 1.$$

Barrier:

$$f(x) = -\sum_i \log(b_i - \langle a_i, x\rangle) - \log(1 - \|x\|_2^2).$$

$$\nu_f = m + 1.$$

Positive Orthant

$$x \geq 0 \quad \Rightarrow \quad f(x) = -\sum_i \log x_i, \quad \nu_f = n.$$

PSD Cone

$$X \succeq 0 \quad \Rightarrow \quad f(X) = -\ln\det X, \quad \nu_f = n.$$

If also entrywise positive:

$$f(X) = -\ln\det X - \sum_{i,j} \log X_{ij}, \quad \nu_f = n + n^2.$$

4.4.6 Interior Point Method

Solve:

$$\min\langle c, x\rangle \quad \text{s.t. } x \in \overline{dom f}.$$

Define:

$$f_\eta(x) = \eta\langle c, x\rangle + f(x).$$

Newton step:

$$n_\eta(x) = -H(x)^{-1}(\eta c + g(x)) = n(x) - \eta c.$$

Short-Step Method

Initialize:

$$\|n_{\eta_0}(x_0)\|_{x_0} \leq \frac{1}{9}.$$

Update:

$$\eta_{k+1} = \left(1 + \frac{1}{8\sqrt{\nu_f}}\right)\eta_k, \quad x_{k+1} = x_k + n_{\eta_{k+1}}(x_k).$$

Theorem 4.4.12. *If*

$$\|n_{\eta_k}(x_k)\|_{x_k} \leq \frac{1}{9},$$

then:

$$\|n_{\eta_{k+1}}(x_{k+1})\|_{x_{k+1}} \leq \frac{1}{9}.$$

Proof. Decompose:

$$n_{\eta_{k+1}}(x_k) = \frac{\eta_{k+1}}{\eta_k} n_{\eta_k}(x_k) + \left(1 - \frac{\eta_{k+1}}{\eta_k}\right) n(x_k),$$

and use SC bounds. □

Complexity

To reach $\eta_k \sim 1/\varepsilon$:

$$k = O(\sqrt{\nu_f}\log(1/\varepsilon)).$$

Alternative (Backward View)

Let $\tilde{c} = g(x)$.

Then x minimizes:

$$\min\langle \tilde{c}, x\rangle + f(x),$$

i.e., central path reversal.

Long-Step / Predictor-Corrector

Practical improvements:

- Long-step: increase η aggressively
- Multiple Newton corrections
- Predictor-corrector: jump + recenter

Worked Example: Central Path on an Interval

For

$$\min_{0<x<1} -x, \tag{4.4.1}$$

the logarithmic-barrier subproblem with parameter $t > 0$ is

$$\min_{0<x<1} -tx - \log x - \log(1-x). \tag{4.4.2}$$

Stationarity gives

$$-t - \frac{1}{x} + \frac{1}{1-x} = 0 \quad \Longleftrightarrow \quad tx^2 + (2-t)x - 1 = 0. \tag{4.4.3}$$

The root in $(0,1)$ is

$$x(t) = \frac{t - 2 + \sqrt{t^2+4}}{2t}. \tag{4.4.4}$$

As $t \to \infty$, $x(t) = 1 - t^{-1} + O(t^{-2})$, so the central path approaches the boundary optimum from the strict interior. The approximate inequality multiplier associated with $x \leq 1$ is $\lambda(t) = 1/[t(1-x(t))] \to 1$, matching the KKT multiplier of the original problem.

4.4.7 Primal-Dual View

Barrier problem:

$$\min f(x) + \langle c, x\rangle + B(x),$$

with:

$$\nabla B(x) = \sum_i \nabla g_i(x) \cdot \frac{1}{-g_i(x)}.$$

KKT-like system:

$$\begin{cases} \nabla f(x) + \nabla g(x)^T u + A^T y = 0 \\ u_i g_i(x) = -\frac{1}{\eta} \\ Ax = b \\ u_i > 0,\ g_i(x) < 0 \end{cases}$$

Thus interior-point methods can be viewed as:

Newton applied to perturbed KKT conditions.

Chapter 5

Constrained Minimax Optimization

5.1 Problem Formulation and Saddle Points

Minimax optimization studies problems of the form

$$\inf_{x\in X}\sup_{y\in Y}\Phi(x,y), \tag{5.1.1}$$

where $X\subseteq\mathbb{R}^n$ and $Y\subseteq\mathbb{R}^m$ encode constraints on two interacting decision variables. The variable x is chosen by a minimizing player and y by a maximizing player. This language is useful even when no literal game is present: Lagrange multipliers, adversarial perturbations, and uncertain model parameters all generate minimax structure.

The reversed problem is

$$\sup_{y\in Y}\inf_{x\in X}\Phi(x,y). \tag{5.1.2}$$

The two values need not agree. For every function and every pair of nonempty sets, however,

$$\sup_{y\in Y}\inf_{x\in X}\Phi(x,y)\le\inf_{x\in X}\sup_{y\in Y}\Phi(x,y). \tag{5.1.3}$$

Indeed, $\inf_{u\in X}\Phi(u,y)\le\Phi(x,y)\le\sup_{v\in Y}\Phi(x,v)$ for every (x,y), and taking the outer supremum and infimum proves the claim. The difference between the right and left sides is the *duality gap*.

Definition 5.1.1 (Saddle point)**.** A feasible pair $(x^\star,y^\star)\in X\times Y$ is a saddle point of Φ if

$$\Phi(x^\star,y)\le\Phi(x^\star,y^\star)\le\Phi(x,y^\star)\qquad\text{for every } x\in X,\ y\in Y. \tag{5.1.4}$$

Thus $x^\star$ minimizes $\Phi(\cdot,y^\star)$ on X, while $y^\star$ maximizes $\Phi(x^\star,\cdot)$ on Y.

If a saddle point exists, then the minimax and maximin values agree and equal $\Phi(x^\star,y^\star)$. The converse requires attainment: equality of the two values alone need not produce a feasible saddle point on noncompact sets.

Define the upper and lower value functions

$$p(x) = \sup_{y \in Y} \Phi(x, y), \qquad d(y) = \inf_{x \in X} \Phi(x, y). \tag{5.1.5}$$

The primal problem minimizes p, whereas the dual problem maximizes d. For a candidate (x, y), the saddle residual

$$\mathcal{G}(x, y) = \sup_{v \in Y} \Phi(x, v) - \inf_{u \in X} \Phi(u, y) = p(x) - d(y) \tag{5.1.6}$$

is nonnegative and vanishes exactly at a saddle point when both extrema are attained. Unlike a gradient norm, $\mathcal{G}$ measures both optimality and the interaction between the players.

Example 5.1.1 (Failure of minimax equality)**.** Let $X = Y = \{-1, 1\}$ and $\Phi(x, y) = xy$. For each x, the maximizing player chooses $y = x$, so $\min_x \max_y xy = 1$. For each y, the minimizing player chooses $x = -y$, so $\max_y \min_x xy = -1$. The gap is caused by the nonconvex pure-strategy sets. Replacing them by their convex hulls $[-1, 1]$ permits the saddle point $(0, 0)$ and makes both values zero.

5.2 Minimax Theorems

Convexity in the minimizing variable and concavity in the maximizing variable are the fundamental structural assumptions. A function Φ is *convex–concave* on $X \times Y$ if $\Phi(\cdot, y)$ is convex on X for each y and $\Phi(x, \cdot)$ is concave on Y for each x.

Theorem 5.2.1 (Sion's minimax theorem)**.** *Let X be a compact convex subset of a linear topological space and let Y be convex. Suppose $\Phi(\cdot, y)$ is lower semicontinuous and convex on X for every $y \in Y$, while $\Phi(x, \cdot)$ is upper semicontinuous and concave on Y for every $x \in X$. Then*

$$\min_{x \in X} \sup_{y \in Y} \Phi(x, y) = \sup_{y \in Y} \min_{x \in X} \Phi(x, y). \tag{5.2.1}$$

If Y is also compact, both outer extrema are attained and a saddle point exists.

Proof idea. For a scalar a strictly above the maximin value, the convex lower level sets $C_y = \{x \in X : \Phi(x, y) \le a\}$ have the finite intersection property. Convexity and a separating-hyperplane argument establish this property; compactness of X then gives $\bigcap_{y \in Y} C_y \neq \varnothing$. Hence some x satisfies $\sup_y \Phi(x, y) \le a$. Letting a decrease to the maximin value gives the reverse inequality to (5.1.3). □

Compactness can be replaced by coercivity or by compactness of one appropriate level set. These variants matter for constrained optimization because a multiplier set is often unbounded. One must then prove that optimal multipliers lie in a bounded subset before applying a compact minimax theorem.

Example 5.2.1 (Finite zero-sum matrix game)**.** Let $A \in \mathbb{R}^{n \times m}$ and let Δ_n, Δ_m be probability simplices. The expected payoff is $\Phi(x, y) = x^T A y$. Since this function is bilinear and both simplices are compact and convex,

$$\min_{x \in \Delta_n} \max_{y \in \Delta_m} x^T A y = \max_{y \in \Delta_m} \min_{x \in \Delta_n} x^T A y. \tag{5.2.2}$$

The vectors x and y are mixed strategies. The theorem explains why randomization closes the pure-strategy gap seen in the preceding example.

5.3 Lagrangian Duality as a Minimax Problem

Consider the constrained program

$$\min_{x\in X} f(x) \quad \text{subject to} \quad g_i(x) \le 0 \ (i = 1, \dots, q), \qquad Ax = b, \tag{5.3.1}$$

where X is a closed convex set. Its Lagrangian is

$$L(x, \lambda, \nu) = f(x) + \lambda^T g(x) + \nu^T (Ax - b), \qquad \lambda \in \mathbb{R}^q_+, \nu \in \mathbb{R}^r. \tag{5.3.2}$$

For a fixed x, maximizing over the multipliers returns $f(x)$ when x is feasible and $+\infty$ otherwise. Therefore the primal value is exactly

$$p^\star = \inf_{x\in X} \sup_{\lambda\ge 0,\nu} L(x, \lambda, \nu). \tag{5.3.3}$$

Reversing the order produces the Lagrange dual value

$$d^\star = \sup_{\lambda\ge 0,\nu} \inf_{x\in X} L(x, \lambda, \nu). \tag{5.3.4}$$

Thus weak duality is precisely the general minimax inequality. Strong duality is a minimax equality for the Lagrangian.

Theorem 5.3.1 (Slater and saddle points)**.** *Suppose f and g_i are proper closed convex functions, the equality constraints are affine, and there is a relative-interior point $\bar{x} \in X$ such that $g_i(\bar{x}) < 0$ and $A\bar{x} = b$. If the primal optimum is attained, then there exist multipliers $(\lambda^\star, \nu^\star)$ such that $(x^\star, \lambda^\star, \nu^\star)$ is a saddle point of the Lagrangian:*

$$L(x^\star, \lambda, \nu) \le L(x^\star, \lambda^\star, \nu^\star) \le L(x, \lambda^\star, \nu^\star) \tag{5.3.5}$$

for all $x \in X$, $\lambda \ge 0$, and $\nu \in \mathbb{R}^r$.

When the functions are differentiable, the saddle conditions are equivalent to the KKT system

$$0 \in \nabla f(x^\star) + \sum_{i=1}^{q} \lambda_i^\star \nabla g_i(x^\star) + A^T \nu^\star + N_X(x^\star), \tag{5.3.6}$$

$$g(x^\star) \le 0, \qquad Ax^\star = b, \tag{5.3.7}$$

$$\lambda^\star \ge 0, \qquad \lambda_i^\star g_i(x^\star) = 0 \quad (i = 1, \dots, q). \tag{5.3.8}$$

Here $N_X(x)$ is the normal cone of X. This formulation makes clear that multiplier ascent is not an auxiliary trick: it is the maximizing half of a constrained saddle problem.

Example 5.3.1 (Equality-constrained quadratic)**.** For $Q \succ 0$, consider

$$\min_x \frac{1}{2} x^T Q x + c^T x \quad \text{subject to} \quad Ax = b. \tag{5.3.9}$$

The Lagrangian $L(x,\nu) = \frac{1}{2}x^TQx + c^Tx + \nu^T(Ax-b)$ is strongly convex in x and affine in ν. Its saddle equations are

$$\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ \nu^\star \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix}. \tag{5.3.10}$$

If A has full row rank, eliminating x gives $\nu^\star = -(AQ^{-1}A^T)^{-1}(b + AQ^{-1}c)$ and $x^\star = -Q^{-1}(c + A^T\nu^\star)$.

5.4 First-Order Conditions and Variational Inequalities

Assume Φ is differentiable and X, Y are closed convex sets. Define the saddle operator

$$F(x,y) = \begin{bmatrix} \nabla_x \Phi(x,y) \\ -\nabla_y \Phi(x,y) \end{bmatrix}, \qquad Z = X \times Y. \tag{5.4.1}$$

The minus sign converts maximization in y into a common monotone-inclusion convention.

Proposition 5.4.1 (Variational-inequality characterization)**.** *If Φ is convex–concave, then $z^\star = (x^\star, y^\star)$ is a saddle point if and only if*

$$\langle F(z^\star), z - z^\star \rangle \geq 0 \qquad \text{for every } z \in Z. \tag{5.4.2}$$

Equivalently, $0 \in F(z^\star) + N_Z(z^\star)$.

Proof. The first-order condition for minimizing $\Phi(\cdot, y^\star)$ on X is $\langle \nabla_x \Phi(x^\star, y^\star), x - x^\star \rangle \geq 0$. The first-order condition for maximizing $\Phi(x^\star, \cdot)$ on Y is $\langle -\nabla_y \Phi(x^\star, y^\star), y - y^\star \rangle \geq 0$. Adding them gives (5.4.2). Convexity and concavity make the two first-order conditions sufficient. □

Proposition 5.4.2 (Monotonicity)**.** *If Φ is differentiable and convex–concave, then the operator F in (5.4.1) is monotone:*

$$\langle F(z) - F(z'), z - z' \rangle \geq 0 \qquad (z, z' \in Z). \tag{5.4.3}$$

If $\Phi(\cdot, y)$ is μ_x-strongly convex and $\Phi(x, \cdot)$ is μ_y-strongly concave, then F is strongly monotone with modulus $\min\{\mu_x, \mu_y\}$.

For twice differentiable Φ, the symmetric part of the Jacobian of F is block diagonal:

$$\frac{\nabla F(z) + \nabla F(z)^T}{2} = \begin{bmatrix} \nabla^2_{xx}\Phi(z) & 0 \\ 0 & -\nabla^2_{yy}\Phi(z) \end{bmatrix}. \tag{5.4.4}$$

The mixed Hessian creates a skew-symmetric rotational component. This rotation is why algorithms that work for ordinary minimization can oscillate on saddle problems even when F is monotone.

5.5 Projected Gradient Descent–Ascent

The simultaneous projected gradient descent–ascent iteration is

$$x_{k+1} = \Pi_X\{x_k - \eta_k \nabla_x \Phi(x_k, y_k)\}, \tag{5.5.1}$$

$$y_{k+1} = \Pi_Y\{y_k + \eta_k \nabla_y \Phi(x_k, y_k)\}. \tag{5.5.2}$$

Equivalently, $z_{k+1} = \Pi_Z\{z_k - \eta_k F(z_k)\}$. Projection incorporates the hard constraints exactly at every iteration.

Algorithm 1 Projected Gradient Descent–Ascent

Require: Closed convex sets X, Y, initial (x_0, y_0), step sizes $\{\eta_k\}$, iteration count T

1: **for** $k = 0, \dots, T-1$ **do**
2: $\quad g_k^x \leftarrow \nabla_x \Phi(x_k, y_k)$ and $g_k^y \leftarrow \nabla_y \Phi(x_k, y_k)$
3: $\quad x_{k+1} \leftarrow \Pi_X(x_k - \eta_k g_k^x)$
4: $\quad y_{k+1} \leftarrow \Pi_Y(y_k + \eta_k g_k^y)$
5: **end for**
6: **return** the weighted averages $\bar{x}_T = (\sum_k \eta_k x_k)/(\sum_k \eta_k)$ and $\bar{y}_T = (\sum_k \eta_k y_k)/(\sum_k \eta_k)$

The algorithm returns averages in the merely convex–concave regime because the last iterates may rotate. In a strongly convex–concave problem, the last iterate is meaningful and averaging is optional.

Example 5.5.1 (Cycling in a bilinear game)**.** For $\Phi(x, y) = xy$ on $\mathbb{R}^2$, unconstrained simultaneous GDA gives

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & -\eta \\ \eta & 1 \end{bmatrix} \begin{bmatrix} x_k \\ y_k \end{bmatrix}. \tag{5.5.3}$$

The multiplier has modulus $\sqrt{1+\eta^2} > 1$, so the iterates spiral away from the saddle $(0, 0)$. Alternating the two updates may prevent immediate expansion for some step sizes but does not remove the underlying rotational difficulty.

For compact Z and bounded $\|F(z)\| \le G$, projected GDA with diminishing steps admits an ergodic guarantee. Let

$$\bar{x}_T = \frac{\sum_{k=0}^{T-1} \eta_k x_k}{\sum_{k=0}^{T-1} \eta_k}, \qquad \bar{y}_T = \frac{\sum_{k=0}^{T-1} \eta_k y_k}{\sum_{k=0}^{T-1} \eta_k}. \tag{5.5.4}$$

If D bounds the diameter of Z, nonexpansiveness of projection gives

$$\sum_{k=0}^{T-1} \eta_k \langle F(z_k), z_k - z \rangle \le \frac{D^2 + G^2 \sum_{k=0}^{T-1} \eta_k^2}{2} \qquad (z \in Z). \tag{5.5.5}$$

Convexity–concavity then yields

$$\mathcal{G}(\bar{x}_T, \bar{y}_T) \le \frac{D^2 + G^2 \sum_{k=0}^{T-1} \eta_k^2}{2 \sum_{k=0}^{T-1} \eta_k}. \tag{5.5.6}$$

With $\eta_k = D/(G\sqrt{T})$, the gap is $O(DG/\sqrt{T})$. This is an ergodic result: the averages can converge even while the last iterates cycle.

5.6 Extragradient and Mirror–Prox

The extragradient method evaluates the operator at a predictive point before making the actual update:

$$\widehat{z}_k = \Pi_Z\{z_k - \eta F(z_k)\}, \tag{5.6.1}$$
$$z_{k+1} = \Pi_Z\{z_k - \eta F(\widehat{z}_k)\}. \tag{5.6.2}$$

The second step starts again from z_k but uses the look-ahead gradient. This correction anticipates the rotational component of F.

Algorithm 2 Projected Extragradient for a Constrained Saddle Problem

Require: Operator F, feasible set $Z = X \times Y$, initial $z_0 \in Z$, step size η, iteration count T
1: **for** $k = 0, \dots, T-1$ **do**
2: $\quad \widehat{z}_k \leftarrow \Pi_Z(z_k - \eta F(z_k))$
3: $\quad z_{k+1} \leftarrow \Pi_Z(z_k - \eta F(\widehat{z}_k))$
4: **end for**
5: **return** $\bar{z}_T = T^{-1} \sum_{k=0}^{T-1} \widehat{z}_k$

Each iteration uses two operator evaluations and two projections. This extra cost buys the $O(1/T)$ monotone gap rate, compared with the generic $O(1/\sqrt{T})$ rate of one-call projected GDA.

Theorem 5.6.1 (Extragradient gap bound)**.** *Suppose Z is closed, convex, and has diameter at most D. Let F be monotone and L-Lipschitz on Z. With $0 < \eta \le 1/(2L)$, the weighted average $\bar{z}_T = T^{-1} \sum_{k=0}^{T-1} \widehat{z}_k$ satisfies the weak variational-inequality gap bound*

$$\sup_{z \in Z} \langle F(z), \bar{z}_T - z \rangle \le \frac{D^2}{2\eta T}. \tag{5.6.3}$$

For a convex–concave saddle problem this implies $\mathcal{G}(\bar{x}_T, \bar{y}_T) = O(LD^2/T)$.

Main inequality. Projection optimality, applied once to each line of (5.6.2), gives for every $z \in Z$

$$2\eta \langle F(\widehat{z}_k), \widehat{z}_k - z \rangle \le \|z_k - z\|^2 - \|z_{k+1} - z\|^2 \tag{5.6.4}$$
$$- \|z_k - \widehat{z}_k\|^2 - \|z_{k+1} - \widehat{z}_k\|^2 \tag{5.6.5}$$
$$+ 2\eta \langle F(\widehat{z}_k) - F(z_k), \widehat{z}_k - z_{k+1} \rangle . \tag{5.6.6}$$

Lipschitz continuity and Young's inequality absorb the last term into the two negative squared norms when $\eta \le 1/(2L)$. Summing telescopes the distance terms. Monotonicity replaces $F(\widehat{z}_k)$ by $F(z)$, and Jensen's inequality gives (5.6.3). □

For non-Euclidean constraints, projection may be replaced by a Bregman prox map. Let h be strongly convex on Z and $D_h(u,z) = h(u) - h(z) - \langle \nabla h(z), u - z \rangle$. Mirror–Prox performs two updates of the form

$$\mathrm{Prox}_z^h(\eta g) = \arg\min_{u \in Z} \{\eta \langle g, u \rangle + D_h(u,z)\}. \tag{5.6.7}$$

Euclidean extragradient corresponds to $h(z) = \frac{1}{2}\|z\|^2$. Negative entropy is often preferable on a simplex because its prox map is a multiplicative-weights update and introduces only logarithmic dependence on the dimension.

Algorithm 3 Mirror–Prox

Require: Distance generator h, initial $z_0 \in Z$, step size η

1: **for** $k = 0, \dots, T-1$ **do**
2: $\quad \widehat{z}_k \leftarrow \mathrm{Prox}_{z_k}^h(\eta F(z_k))$
3: $\quad z_{k+1} \leftarrow \mathrm{Prox}_{z_k}^h(\eta F(\widehat{z}_k))$
4: **end for**
5: **return** the average of the predictive points $\widehat{z}_k$

5.6.1 Optimistic Gradient Descent–Ascent

Optimistic GDA uses the current gradient together with a correction based on the preceding gradient:

$$z_{k+1} = \Pi_Z\{z_k - 2\eta F(z_k) + \eta F(z_{k-1})\}. \tag{5.6.8}$$

It uses one new operator evaluation per iteration after initialization, whereas extragradient uses two. The correction extrapolates the local rotation of the saddle dynamics. For unconstrained bilinear games, OGDA is closely related to extragradient after a change of state variables and converges for sufficiently small step size.

Algorithm 4 Optimistic Gradient Descent–Ascent

Require: Initial $z_{-1} = z_0 \in Z$, step size η

1: Evaluate $g_{-1} = g_0 = F(z_0)$
2: **for** $k = 0, \dots, T-1$ **do**
3: $\quad z_{k+1} \leftarrow \Pi_Z(z_k - 2\eta g_k + \eta g_{k-1})$
4: $\quad g_{k+1} \leftarrow F(z_{k+1})$
5: **end for**
6: **return** an ergodic average or the last iterate under stronger assumptions

Example 5.6.1 (OGDA on the scalar bilinear game)**.** For $\Phi(x,y) = xy$, $F(x,y) = (y, -x)$. The method becomes

$$x_{k+1} = x_k - 2\eta y_k + \eta y_{k-1}, \tag{5.6.9}$$

$$y_{k+1} = y_k + 2\eta x_k - \eta x_{k-1}. \tag{5.6.10}$$

Unlike simultaneous GDA, the augmented two-step recurrence contains damping. For sufficiently small η, its nontrivial characteristic roots lie inside the unit circle, so the last iterates converge to the origin rather than spiraling outward.

5.6.2 Stochastic Saddle-Point Methods

Suppose $F(z) = \mathbb{E}[F(z;\xi)]$ and only unbiased stochastic estimates are available. Stochastic projected GDA uses

$$z_{k+1} = \Pi_Z\{z_k - \eta_k F(z_k;\xi_k)\}. \tag{5.6.11}$$

If $\mathbb{E}[\|F(z;\xi)\|^2] \le G^2$, the same projection argument as in the deterministic method yields an expected $O(1/\sqrt{T})$ ergodic gap. A stochastic extragradient step should normally use independent samples in its prediction and correction evaluations; reusing the same sample can introduce a correlation term that invalidates the usual unbiased proof.

Variance reduction becomes useful when $F(z) = N^{-1}\sum_{i=1}^{N} F_i(z)$. A snapshot full gradient can be combined with $F_i(z) - F_i(\tilde{z})$ to form an unbiased estimator whose variance shrinks near the snapshot. In strongly monotone finite-sum problems this can replace the noise-limited sublinear rate by geometric convergence up to the cost of periodic full passes.

Algorithm 5 Stochastic Extragradient

Require: Initial $z_0 \in Z$, steps η_k

1: **for** $k = 0, \dots, T-1$ **do**
2: Draw ξ_k and set $\widehat{z}_k \leftarrow \Pi_Z\{z_k - \eta_k F(z_k;\xi_k)\}$
3: Draw an independent ξ_k' and set $z_{k+1} \leftarrow \Pi_Z\{z_k - \eta_k F(\widehat{z}_k;\xi_k')\}$
4: **end for**
5: **return** the step-size-weighted average of $\widehat{z}_k$

5.6.3 Proximal Best Responses

When each block can be optimized more accurately than a single gradient step, one may use regularized best responses:

$$x_{k+1} \in \arg\min_{x\in X}\left\{\Phi(x, y_k) + \frac{1}{2\eta}\|x - x_k\|^2\right\}, \tag{5.6.12}$$

$$y_{k+1} \in \arg\max_{y\in Y}\left\{\Phi(x_{k+1}, y) - \frac{1}{2\eta}\|y - y_k\|^2\right\}. \tag{5.6.13}$$

The proximal terms stabilize the interaction and make each subproblem strongly convex or strongly concave when the original block has only weak curvature. Exact best responses can nevertheless be expensive; inexact solves require a summable or rate-compatible error condition.

5.7 Strong Convexity–Concavity and Linear Convergence

If F is μ-strongly monotone and L-Lipschitz, the saddle point is unique. The projected forward step satisfies

$$\|z_{k+1} - z^\star\|^2 \le \|z_k - z^\star - \eta(F(z_k) - F(z^\star))\|^2 \tag{5.7.1}$$

$$\le (1 - 2\eta\mu + \eta^2 L^2)\,\|z_k - z^\star\|^2 . \tag{5.7.2}$$

Thus any $0 < \eta < 2\mu/L^2$ gives linear convergence. This conservative range reflects the skew component of a general saddle operator. Extragradient and proximal-point methods allow step sizes of order $1/L$ and obtain the more favorable condition-number dependence expected for monotone inclusions.

The proximal-point saddle update is

$$0 \in F(z_{k+1}) + N_Z(z_{k+1}) + \frac{1}{\eta}(z_{k+1} - z_k). \tag{5.7.3}$$

It is implicit but firmly nonexpansive in the monotone case. Extragradient may be viewed as an explicit approximation to this resolvent step.

5.8 Primal–Dual Algorithms for Constrained Problems

Consider the composite constrained problem

$$\min_x f(x) + g(Kx), \tag{5.8.1}$$

where f and g are proper closed convex functions and K is linear. Since $g(Kx) = \sup_y\{\langle Kx, y\rangle - g^*(y)\}$, the problem is equivalent to

$$\min_x \max_y \Phi(x, y) := f(x) + \langle Kx, y\rangle - g^*(y). \tag{5.8.2}$$

The primal–dual hybrid gradient method updates

$$y_{k+1} = \operatorname{prox}_{\sigma g^*}\{y_k + \sigma K\bar{x}_k\}, \tag{5.8.3}$$

$$x_{k+1} = \operatorname{prox}_{\tau f}\{x_k - \tau K^T y_{k+1}\}, \tag{5.8.4}$$

$$\bar{x}_{k+1} = x_{k+1} + \theta(x_{k+1} - x_k). \tag{5.8.5}$$

For $\theta = 1$ and $\tau\sigma\,\|K\|^2 < 1$, the iterates converge to a saddle point under standard existence assumptions, and ergodic primal–dual gaps decay as $O(1/T)$.

Example 5.8.1 (Basis-pursuit denoising)**.** The problem

$$\min_x \|x\|_1 \quad \text{subject to} \quad \|Ax - b\|_2 \le \delta \tag{5.8.6}$$

has the form (5.8.1) with $f(x) = \|x\|_1$ and $g(u) = \iota_{B(b,\delta)}(u)$. The dual proximal map can be computed from projection onto the Euclidean ball by Moreau's identity, while the primal proximal map is soft thresholding. PDHG therefore handles the nonsmooth objective and the hard data-fidelity constraint without forming normal equations.

For inequality constraints $g(x) \le 0$, projected primal–dual gradient takes the form

$$x_{k+1} = \Pi_X\{x_k - \eta_k[\nabla f(x_k) + Jg(x_k)^T\lambda_k]\}, \tag{5.8.7}$$
$$\lambda_{k+1} = \Pi_{\mathbb{R}^q_+}\{\lambda_k + \eta_k g(x_k)\}. \tag{5.8.8}$$

The dual update increases multipliers on violated constraints. Without a bound on the multiplier sequence, a compact-domain gap theorem cannot be applied directly. Slater's condition supplies such a bound near optimality, or one may explicitly project onto a ball known to contain an optimal multiplier.

5.9 Nonconvex–Concave Minimax Problems

Many modern models are nonconvex in x but remain concave in y. Define

$$p(x) = \max_{y\in Y} \Phi(x, y). \tag{5.9.1}$$

Even if Φ is smooth, p may be nonsmooth when the maximizer is not unique.

Theorem 5.9.1 (Danskin's theorem)**.** *Let Y be compact and let Φ and $\nabla_x\Phi$ be continuous. Define $Y^\star(x) = \arg\max_{y\in Y} \Phi(x, y)$. Then the directional derivative is*

$$p'(x; d) = \max_{y\in Y^\star(x)} \langle \nabla_x\Phi(x, y), d\rangle . \tag{5.9.2}$$

If $Y^\star(x) = \{y^\star(x)\}$, then p is differentiable and $\nabla p(x) = \nabla_x\Phi(x, y^\star(x))$.

If $\Phi(x, \cdot)$ is μ-strongly concave and its full gradient is L-Lipschitz, the maximizer is unique and Lipschitz:

$$\|y^\star(x) - y^\star(x')\| \le \frac{L}{\mu} \|x - x'\| . \tag{5.9.3}$$

Consequently the value function is smooth, with a Lipschitz constant depending on L and the condition number L/μ. A two-timescale method approximately maximizes in y through several ascent steps before taking a descent step in x. Its natural target is stationarity of the constrained value function, for example

$$\mathcal{R}_\eta(x) = \frac{1}{\eta} [x - \Pi_X\{x - \eta\nabla p(x)\}] , \tag{5.9.4}$$

rather than global saddle optimality, which is generally intractable in the nonconvex setting.

When the inner problem is merely concave, p can be nonsmooth and the choice of an arbitrary best response supplies only a subgradient. Regularization, such as replacing $\Phi(x, y)$ by $\Phi(x, y) - \frac{\mu}{2}\|y\|^2$, produces a smooth approximation but biases the original minimax problem. The regularization error and optimization error must be controlled together.

5.10 Robust Optimization

5.10.1 Uncertain Constraints and Robust Counterparts

A deterministic model treats its coefficients as known. Robust optimization instead specifies an uncertainty set $\mathcal{U}$ and requires feasibility for every realization in that set. The robust counterpart of

$$f_i(x,u) \le 0, \qquad u \in \mathcal{U}_i, \tag{5.10.1}$$

is the single worst-case constraint

$$\sup_{u\in\mathcal{U}_i} f_i(x,u) \le 0. \tag{5.10.2}$$

This is a minimax construction. If $f_i(\cdot,u)$ is convex for every u, then its pointwise supremum is convex, even when the maximizing uncertainty depends nonsmoothly on x.

The uncertainty set encodes the desired protection. A box permits every coefficient to move independently to an extreme value and can be overly conservative. Ellipsoids couple deviations through a quadratic budget. A polyhedral set permits correlated but linearly constrained deviations and typically yields an LP robust counterpart.

Theorem 5.10.1 (Robust linear inequality)**.** *Suppose the uncertain coefficient has the affine representation $a(u) = a_0 + Pu$ and $\mathcal{U} = \{u : \|u\| \le \rho\}$. Then*

$$a(u)^T x \le b \quad \textit{for every } u \in \mathcal{U} \tag{5.10.3}$$

is equivalent to

$$a_0^T x + \rho \left\| P^T x \right\|_* \le b, \tag{5.10.4}$$

where $\|\cdot\|_*$ is the dual norm.

Proof. The largest left side is

$$\sup_{\|u\|\le\rho} (a_0 + Pu)^T x = a_0^T x + \rho \sup_{\|v\|\le 1} \left\langle v, P^T x \right\rangle = a_0^T x + \rho \left\| P^T x \right\|_* . \tag{5.10.5}$$

Substitution into (5.10.2) gives the claim. □

For Euclidean uncertainty, (5.10.4) is a second-order cone constraint. For ℓ_∞ uncertainty it adds an ℓ_1 term, while ℓ_1 uncertainty produces an ℓ_∞ term. This is the basic mechanism connecting robust optimization and regularization.

Example 5.10.1 (Robust resource constraint)**.** Suppose nominal resource use is $\bar{a} = (2,1)$ and coefficient error is $a = \bar{a} + u$ with $\|u\|_2 \le 0.2$. Requiring $a^T x \le 5$ for every error gives

$$2x_1 + x_2 + 0.2\sqrt{x_1^2 + x_2^2} \le 5. \tag{5.10.6}$$

At $x = (2,1)$ the nominal use is 5, but the robust left side is $5 + 0.2\sqrt{5} > 5$, so the nominally active solution is not robust feasible. The worst perturbation is $u^\star = 0.2x/\|x\|_2$.

5.10.2 Polyhedral Uncertainty and LP Duality

Let $\mathcal{U} = \{u : Du \le d\}$ and consider

$$\sup_{u \in \mathcal{U}} (a_0 + Pu)^T x \le b. \tag{5.10.7}$$

The inner LP has objective $(P^T x)^T u$. Its dual is

$$\inf_{q \ge 0} \; d^T q \quad \text{subject to} \quad D^T q = P^T x. \tag{5.10.8}$$

Assuming the uncertainty set is nonempty and the support function is finite, strong LP duality shows that the robust constraint is equivalent to the existence of $q \ge 0$ satisfying

$$D^T q = P^T x, \qquad a_0^T x + d^T q \le b. \tag{5.10.9}$$

Thus a semi-infinite constraint becomes finitely many linear constraints by introducing dual variables. Robust counterparts should be derived in this way rather than by sampling a few uncertainty realizations, which gives no worst-case guarantee.

5.10.3 Robust Least Squares

Consider matrix uncertainty $A + \Delta$ with $\|\Delta\|_F \le \rho$ and

$$\min_x \max_{\|\Delta\|_F \le \rho} \|(A + \Delta)x - b\|_2 . \tag{5.10.10}$$

For this Frobenius uncertainty, the inner maximum is

$$\|Ax - b\|_2 + \rho \|x\|_2 . \tag{5.10.11}$$

The upper bound follows from the triangle inequality and $\|\Delta x\|_2 \le \rho \|x\|_2$. Equality is obtained by a rank-one perturbation mapping x in the direction of the residual. Hence robust least squares is equivalent to a norm-regularized residual problem. Squaring (5.10.11) is allowed because both sides are nonnegative, but it does not produce ordinary ridge regression; the cross term remains.

Example 5.10.2 (Scalar robust least squares)**.** Let $A = 1$, $b = 2$, and $|\Delta| \le 1/2$. The robust objective is $|x - 2| + |x|/2$. On $0 \le x \le 2$ this equals $2 - x/2$ and is minimized at $x = 2$. For $x \ge 2$ it equals $3x/2 - 2$ and increases, while for $x < 0$ it is $2 - 3x/2$. Thus $x^\star = 2$ with worst-case residual one. The nominal residual is zero, but the robust optimum accepts unavoidable perturbation error.

5.10.4 Adversarially Robust Regression

Let $\ell(w; a, b)$ be a convex loss and let each feature vector be perturbed in a compact convex uncertainty set U_i. Robust empirical risk minimization is

$$\min_{w \in W} \frac{1}{N} \sum_{i=1}^{N} \max_{u_i \in U_i} \ell(w; a_i + u_i, b_i). \tag{5.10.12}$$

For linear regression with squared loss and $U_i = \{u : \|u\|_2 \le \rho\}$,

$$\max_{\|u\|\le\rho} \Big((a_i + u)^T w - b_i\Big)^2 = \Big(\big|a_i^T w - b_i\big| + \rho \|w\|_2\Big)^2 . \tag{5.10.13}$$

The inner maximization therefore yields an explicit convex robust objective. This example also warns that an apparently smooth minimax model can produce a nonsmooth value function through an absolute value.

When no closed form is available, adversarial training approximates the inner maximum by projected ascent and then updates the outer variable. The inner loop must be sufficiently accurate: a weak adversary optimizes a lower bound on the intended robust loss and can give a false impression of robustness.

Algorithm 6 Projected Adversarial Training

Require: Initial parameter w_0, uncertainty sets U_i, outer steps η_k, inner ascent step α, inner iteration count J

for $k = 0, \dots, T-1$ **do**
  Draw a minibatch B_k
  for each $i \in B_k$ **do**
    Initialize $u_i^{(0)} \in U_i$ (optionally with a random start)
    for $j = 0, \dots, J-1$ **do**
      $u_i^{(j+1)} \leftarrow \Pi_{U_i}\{u_i^{(j)} + \alpha \nabla_u \ell(w_k; a_i + u_i^{(j)}, b_i)\}$
    end for
  end for
  $w_{k+1} \leftarrow \Pi_W\{w_k - \eta_k |B_k|^{-1} \sum_{i\in B_k} \nabla_w \ell(w_k; a_i + u_i^{(J)}, b_i)\}$
end for
return w_T

If the inner problem is strongly concave, warm-started ascent can track its maximizer. In a nonconcave adversarial problem, multiple random starts and a separate evaluation adversary are diagnostics rather than proofs of global robustness.

5.10.5 Constraint Generation for Robust Programs

When $\mathcal{U}$ is large but the worst-case oracle

$$u_i(x) \in \arg\max_{u\in\mathcal{U}_i} f_i(x, u) \tag{5.10.14}$$

is tractable, a cutting-plane method solves a sequence of finite scenario problems. Each violated worst-case realization becomes a new constraint.

Algorithm 7 Constraint Generation for Robust Optimization

Require: Initial finite scenario sets $S_i^0 \subset \mathcal{U}_i$, tolerance ε

for $k = 0, 1, 2, \ldots$ **do**
  Solve the master problem $x_k \in \arg\min_{x \in X} f_0(x)$ subject to $f_i(x, u) \le 0$ for every $u \in S_i^k$
  for $i = 1, \ldots, q$ **do**
    $u_i^k \in \arg\max_{u \in \mathcal{U}_i} f_i(x_k, u)$
    $v_i^k \leftarrow f_i(x_k, u_i^k)$
  end for
  if $\max_i v_i^k \le \varepsilon$ **then**
    return x_k as an ε-robust feasible solution
  end if
  $S_i^{k+1} \leftarrow S_i^k \cup \{u_i^k\}$ for every violated constraint
end for

The master objective supplies a lower bound on the robust optimum because it contains only a subset of scenarios. If x_k passes the separation oracle, it is robust feasible and its objective supplies a matching upper bound up to the oracle tolerance. Finite convergence occurs for finite uncertainty sets; compact infinite sets generally require a limiting argument.

5.10.6 Distributionally Robust Optimization

Let P be an unknown data distribution and let $\mathcal{P}$ be an ambiguity set around an empirical distribution. The problem

$$\min_{x \in X} \sup_{Q \in \mathcal{P}} \mathbb{E}_Q[\ell(x; \xi)] \tag{5.10.15}$$

guards against distributional misspecification. The ambiguity set may be defined by moments, a statistical divergence, or an optimal-transport distance. Its radius controls a bias–robustness tradeoff: radius zero returns empirical risk minimization, whereas an excessively large radius can produce a vacuous worst-case model.

For a ϕ-divergence ball around a discrete empirical distribution, write $Q = (q_1, \ldots, q_N)$ and $\widehat{P}_i = 1/N$. The inner problem becomes

$$\max_{q \in \Delta_N} \sum_{i=1}^{N} q_i \ell_i(x) \quad \text{subject to} \quad \frac{1}{N} \sum_{i=1}^{N} \phi(N q_i) \le \rho. \tag{5.10.16}$$

This is a finite concave maximization over probabilities and can be handled by primal–dual or mirror methods. Entropic geometry is natural because the adversarial variable lies on a simplex.

For a 1-Wasserstein ball and an $L(x)$-Lipschitz loss in ξ, Kantorovich duality gives, under the standard regularity assumptions,

$$\sup_{Q : W_1(Q, \widehat{P}_N) \le \rho} \mathbb{E}_Q[\ell(x; \xi)] = \frac{1}{N} \sum_{i=1}^{N} \ell(x; \xi_i) + \rho L(x). \tag{5.10.17}$$

Thus a transport ambiguity set produces a regularizer determined by the sensitivity of the loss to data perturbations.

Example 5.10.3 (Worst-case probability reweighting)**.** Suppose two observed losses are $\ell_1 = 1$ and $\ell_2 = 3$, and restrict the adversarial probability to $q_1 + q_2 = 1$ and $|q_2 - 1/2| \le 0.2$. The expected loss is $q_1 + 3q_2 = 1 + 2q_2$, so the adversary chooses $q_2 = 0.7$ and raises the empirical mean from 2 to 2.4. This small example shows that DRO can act through probability reweighting even when the observed support is fixed.

5.10.7 A Primal–Dual Algorithm for Finite-Scenario Robustness

For convex scenario losses $f_i(x)$, the worst-case problem

$$\min_{x \in X} \max_{p \in \Delta_m} \sum_{i=1}^{m} p_i f_i(x) \tag{5.10.18}$$

has a simplex adversary. Euclidean projection can be used, but exponentiated gradient respects the simplex geometry and avoids a sorting-based projection.

Algorithm 8 Primal–Dual Multiplicative Weights for Finite Scenarios

Require: Initial $x_0 \in X$, $p_{0,i} = 1/m$, primal steps η_k, dual step γ
1: **for** $k = 0, \dots, T-1$ **do**
2: $\quad g_k \leftarrow \sum_{i=1}^{m} p_{k,i} g_{k,i}$ with $g_{k,i} \in \partial f_i(x_k)$
3: $\quad x_{k+1} \leftarrow \Pi_X(x_k - \eta_k g_k)$
4: $\quad \widetilde{p}_{k+1,i} \leftarrow p_{k,i} \exp\{\gamma f_i(x_k)\}$
5: $\quad p_{k+1,i} \leftarrow \widetilde{p}_{k+1,i} / \sum_{j=1}^{m} \widetilde{p}_{k+1,j}$
6: **end for**
7: **return** averaged primal and dual iterates

The adversarial weights increase on scenarios with high current loss. Under bounded subgradients and losses, the standard online-learning regret bounds give an $O(1/\sqrt{T})$ saddle gap; Mirror–Prox improves this to $O(1/T)$ when the smooth saddle operator is Lipschitz.

5.10.8 Chance Constraints and Safe Convex Approximations

A chance constraint requires

$$\mathbb{P}\{a(\xi)^T x > b\} \le \varepsilon. \tag{5.10.19}$$

It is probabilistic rather than worst case and is often nonconvex. If only the mean μ and covariance Σ of $a(\xi)$ are known, the one-sided Chebyshev–Cantelli inequality gives the distributionally robust sufficient condition

$$\mu^T x + sqrt\frac{1-\varepsilon}{\varepsilon} \left\| \Sigma^{1/2} x \right\|_2 \le b. \tag{5.10.20}$$

This SOCP constraint guarantees the desired violation probability for every distribution with those moments. It may be conservative because moment information alone permits heavy-tailed distributions.

Example 5.10.4 (Gaussian versus moment-robust safety factors)**.** For a scalar standard-normal uncertainty and $\varepsilon = 0.05$, the exact one-sided Gaussian quantile is approximately 1.645. The moment-robust factor in (5.10.20) is $\sqrt{0.95/0.05} \approx 4.36$. The larger factor is the price of protecting against every distribution having the same first two moments rather than one known Gaussian law.

5.10.9 Static versus Adjustable Robust Decisions

In a multistage problem, some decisions are made before uncertainty is revealed and others afterward. A static robust model forces every decision to be chosen in advance and may be unnecessarily conservative. An adjustable model separates a first-stage decision x from a recourse decision $y(u)$:

$$\min_{x} \sup_{u \in \mathcal{U}} \left\{ c^T x + q(u)^T y(u) \right\} \tag{5.10.21}$$

subject to constraints that must hold for every u and permit y to depend on the realized uncertainty. Optimizing over arbitrary functions $y(\cdot)$ is generally infinite dimensional. A common approximation restricts recourse to an affine decision rule

$$y(u) = y_0 + Y u. \tag{5.10.22}$$

Substitution reduces the model to a static robust problem in (x, y_0, Y). Affine rules need not be optimal, but they preserve tractability for many polyhedral and ellipsoidal uncertainty sets and dominate a fully static rule when recourse is valuable.

Example 5.10.5 (Inventory with affine recourse)**.** Choose initial inventory x before uncertain demand $u \in [0, 10]$ is observed. Emergency supply $y(u) \geq 0$ is purchased afterward and must satisfy $x + y(u) \geq u$. If y were static, feasibility would require $x + y \geq 10$ even at low demand. With the affine rule $y(u) = \alpha + \beta u$, robust feasibility requires

$$x + \alpha + (\beta - 1)u \geq 0 \qquad \text{for every } u \in [0, 10]. \tag{5.10.23}$$

Because the left side is affine in u, it suffices to enforce the two endpoint constraints $x + \alpha \geq 0$ and $x + \alpha + 10(\beta - 1) \geq 0$, together with nonnegativity of the recourse at the endpoints. The semi-infinite policy constraint has become finitely many linear inequalities.

5.10.10 Scenario Approximation and Validation

When an uncertainty distribution can be sampled but an exact robust counterpart is unavailable, the scenario program enforces constraints on i.i.d. draws $u^1, \ldots, u^N$:

$$\min_{x \in X} f_0(x) \quad \text{subject to} \quad f_i(x, u^j) \leq 0 \quad (i = 1, \ldots, q;\ j = 1, \ldots, N). \tag{5.10.24}$$

This is not a worst-case guarantee over the full support. Under convexity and nondegeneracy, scenario theory bounds the out-of-sample violation probability in terms of the effective decision dimension, sample size, and confidence level. An independent validation sample is still valuable: the training scenarios determine the decision and cannot be reused as an unbiased estimate of its violation probability.

If a candidate violates V of M independent validation scenarios, the empirical violation rate is V/M, but uncertainty in that estimate should be reported through a binomial confidence bound. Observing no violations does not establish zero risk; for example, the usual rule of three gives an approximate 95% upper bound of $3/M$.

5.10.11 Sensitivity to the Robustness Radius

Let $v(\rho)$ denote the optimal value when the uncertainty set has radius ρ. Enlarging the set can only worsen a minimization objective, so $v(\rho)$ is nondecreasing. Where differentiable, its derivative is a shadow price for additional robustness. For the dual-norm counterpart

$$a_0^T x + \rho \left\| P^T x \right\|_* \leq b, \tag{5.10.25}$$

the multiplier on this constraint times $\left\| P^T x \right\|_*$ gives the local value sensitivity, subject to the standard envelope conditions. Plotting nominal performance, worst-case performance, and feasibility margins over a range of ρ is often more informative than reporting a single arbitrarily chosen radius.

5.10.12 Maximum of Convex Functions

For convex differentiable functions $f_1, \ldots, f_m$,

$$\min_{x \in X} \max_{1 \leq i \leq m} f_i(x) = \min_{x \in X} \max_{y \in \Delta_m} \sum_{i=1}^{m} y_i f_i(x). \tag{5.10.26}$$

The simplex variable chooses an active worst-case function. At a saddle point, the support of $y^\star$ lies among active indices $i \in \arg\max_j f_j(x^\star)$. The stationarity condition becomes

$$0 \in \sum_{i=1}^{m} y_i^\star \nabla f_i(x^\star) + N_X(x^\star), \tag{5.10.27}$$

which is exactly the subgradient optimality condition for the pointwise maximum.

Example 5.10.6 (Balancing two quadratic losses)**.** Consider $\min_x \max\{(x-1)^2, (x+2)^2\}$. At the optimum the two losses must be equal; otherwise moving toward the larger-loss center decreases the maximum. Solving $(x-1)^2 = (x+2)^2$ gives $x^\star = -1/2$ and value $9/4$. Their gradients at $x^\star$ are -3 and 3, so stationarity holds with adversarial weights $(1/2, 1/2)$. This explicitly realizes the subgradient of the maximum as a convex combination of active gradients.

5.11 Stopping Criteria and Practical Diagnostics

For compact X and Y, the primal–dual gap (5.1.6) is the preferred certificate because it has direct optimization meaning. Computing it requires solving two best-response problems; approximate best responses still provide an upper bound when their optimization errors are included.

For the variational inequality, the natural residual is

$$r_\eta(z) = \frac{1}{\eta}\left[z - \Pi_Z\{z - \eta F(z)\}\right]. \tag{5.11.1}$$

It vanishes exactly when $0 \in F(z) + N_Z(z)$. In Lagrangian problems, this residual should be reported together with primal feasibility, complementarity, and a dual objective bound. A small stationarity residual alone does not exclude severe constraint violation.

Step sizes must reflect the geometry of both blocks. Poor scaling between primal and dual variables can create artificial oscillation even for a well- conditioned underlying problem. Diagonal or operator preconditioning, separate primal and dual step sizes, and mirror maps adapted to simplex or cone constraints are principled ways to correct this imbalance.